KTU 2024 SCHEME · PCECT403 · S4 ECE

Linear Integrated Circuits Playbook

From "what is inside a triangle?" to PLLs and data converters — one friendly step at a time.

INSIDE

M1	Op-amp internals — differential amplifier, current mirrors, IC 741 parameters
M2	Negative feedback & op-amp applications — amplifiers to sample-and-hold
M3	Oscillators, waveform generators, active filters & IC 723 regulator
M4	IC 555, VCO 566, PLL 565, DACs & ADCs
+	Placement & GATE rapid-fire · formula sheet · night-before summary

★ How to use this playbook

An op-amp looks like a humble triangle with three pins, yet it quietly runs your phone's audio, your ECG machine, your car's sensors and every data converter on the planet. Linear ICs is the subject where analog electronics finally becomes *useful*: you stop fighting transistor bias equations and start building real systems with a handful of resistors and capacitors.

Every topic follows the same rhythm: **plain-words intuition first** (as if you have never met the idea before), then the **proper theory and derivation** at KTU end-sem depth, then **solved examples** that get gradually harder, and finally an **interview corner** with the questions core-electronics companies and GATE keep asking. Read the intuition even when you feel you "already know" the topic — interview answers usually live there.

Two golden rules will carry you through most of Module 2 and 3: with negative feedback, **(1) the two inputs sit at the same voltage** and **(2) no current flows into the inputs**. Whenever you are stuck, come back to these two lines.

Suggested routine: one section a day, cover the solution and try each example yourself first, and use the revision page at the end of each module the night before a test. Every numerical answer in this book has been checked with a quick computer calculation.

The boxes you will see

Key idea — the one thing to remember

Think of it like — an everyday picture

Exam tip — how marks are won

Common mistake — how marks are lost

Interview corner — real placement-style Q&A

Formula — results worth memorising

Notation: $v_d = v_1 - v_2$ is the differential input, A is the open-loop gain, β is either the transistor current gain (Module 1) or the feedback fraction (Modules 2–3) — the context always makes it clear. $V_T \approx 26$ mV at room temperature. Complements are written with a prime, e.g. Q′.

≡ Contents

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01

Op-amp Internals & the IC 741

> *Open the triangle once and every op-amp spec (offset, CMRR, slew rate) starts to make sense — and interviewers love asking "what's inside?"*

SYLLABUS CHECKLIST

- ☐ BJT differential amplifier — DC and AC analysis
- ☐ Differential gain, common-mode gain, CMRR, input resistance
- ☐ Constant current source; current mirrors — basic, Wilson, Widlar
- ☐ Active load and level shifter
- ☐ Block diagram of an op-amp
- ☐ IC 741 — ideal vs practical parameters: offset voltage/current, bias current, slew rate, CMRR, PSRR, GBW
- ☐ Frequency compensation basics
- ☐ Equivalent circuit and voltage transfer curve

≈ 9 hours · CO1 · Analyse the internal stages of an op-amp and interpret its parameters

1.1 The differential amplifier — heart of every op-amp

An ordinary amplifier amplifies *one* signal measured against ground. A **differential amplifier** has two inputs and amplifies only the **difference** between them. Anything that appears equally on both inputs — hum from the mains, noise picked up by long wires, a slow drift in temperature — is (ideally) ignored.

> **THINK OF IT LIKE**
>
> Two kids on a see-saw. If both get heavier by the same amount (common-mode change), the see-saw doesn't tilt. Only when one becomes heavier than the other (a difference) does it tilt. The tilt is the output.

The circuit is two identical transistors Q_1, Q_2 whose emitters are tied together and fed by a common emitter resistor R_E (or better, a current source). Because the total current I_T through R_E is nearly fixed, if Q_1 takes more current Q_2 must take less — the pair "steers" the current according to the difference of the base voltages.

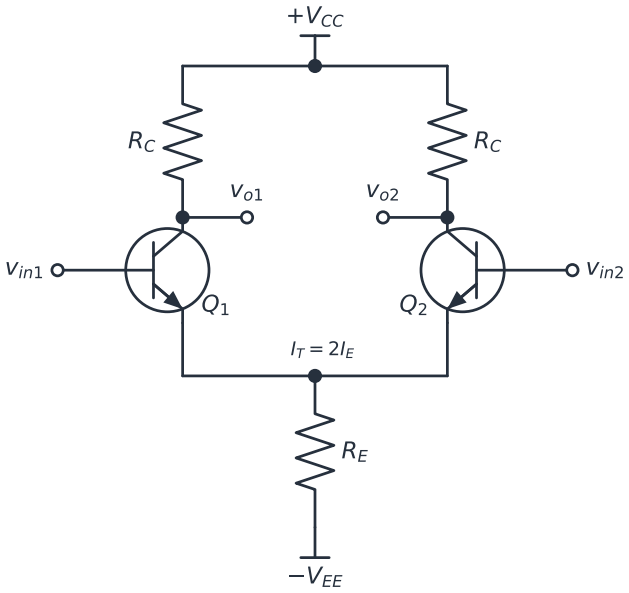


Fig 1.1 Dual-input, balanced-output BJT differential amplifier. Output $v_o = v_{o2} - v_{o1}$.

Depending on how many inputs are used and whether the output is taken between the two collectors (**balanced**) or from one collector to ground (**unbalanced**), we get four configurations:

Configuration	Inputs used	Output taken	A_d	R_i	R_o
Dual-input balanced-output	both	C_1 to C_2	R_C/r_e	$2\beta r_e$	$2R_C$
Dual-input unbalanced-output	both	one collector	$R_C/2r_e$	$2\beta r_e$	R_C
Single-input balanced-output	one (other grounded)	C_1 to C_2	R_C/r_e	$2\beta r_e$	$2R_C$
Single-input unbalanced-output	one	one collector	$R_C/2r_e$	$2\beta r_e$	R_C

Pattern to remember: the gain depends only on the **output** side (unbalanced = half of balanced), and R_i is always $2\beta r_e$ because the signal crosses two base-emitter junctions in series.

1.2 DC analysis of the BJT differential amplifier

DC analysis means finding the operating point (I_C, V_{CE}) with all signals switched off — bases returned to ground through their source resistances R_{in}. Since both halves are identical, analyse just one half.

KVL around the base–emitter loop of Q_1: the base current flows through R_{in}, then V_{BE}, then R_E carries the current of *both* emitters ($2I_E$):

FORMULA — DC OPERATING POINT

$$R_{in}I_B + V_{BE} + 2R_EI_E = V_{EE} \Rightarrow I_E = \frac{V_{EE} - V_{BE}}{2R_E + R_{in}/\beta} \approx \frac{V_{EE} - V_{BE}}{2R_E}$$

$$V_C = V_{CC} - R_CI_C, \quad V_E \approx -V_{BE} \Rightarrow V_{CE} = V_{CC} + V_{BE} - R_CI_C$$

Notice that I_E does **not** depend on R_C — the tail resistor and the negative supply alone set the current. That is why the tail is so important and why we later replace it with a current source.

SOLVED EXAMPLE 1.1 · DC ANALYSIS

A dual-input balanced-output diff amp (Fig 1.1) has $V_{CC} = V_{EE} = 12$ V, $R_C = 5.6$ kΩ, $R_E = 8.2$ kΩ, $\beta = 150$, $V_{BE} = 0.7$ V; source resistances are negligible. Find I_E, V_C and V_{CE}.

- $I_E = (12 - 0.7)/(2 \times 8.2\text{ k}) = 11.3/16.4\text{ k} = 0.689$ mA (each transistor)
- $I_C \approx I_E = 0.689$ mA $\Rightarrow V_C = 12 - 5.6\text{ k} \times 0.689\text{ mA} = 12 - 3.86 = 8.14$ V
- $V_E = -0.7$ V $\Rightarrow V_{CE} = 8.14 - (-0.7) = 8.84$ V

$I_E = I_C \approx 0.689$ mA, $V_C \approx 8.14$ V, $V_{CE} \approx 8.84$ V

1.3 AC analysis: A_d, A_{cm}, CMRR, input resistance

For AC analysis, replace each transistor by its small-signal **r_e model**: a resistance $r_e = V_T/I_E$ in the emitter and a collector current $i_c \approx i_e$. DC supplies become AC ground. Any pair of inputs can be split into two parts:

FORMULA — SPLITTING THE INPUTS

$$v_d = v_{in1} - v_{in2} \quad \text{(differential)} \qquad v_{cm} = \frac{v_{in1} + v_{in2}}{2} \quad \text{(common mode)}$$

$$v_o = A_dv_d + A_{cm}v_{cm}$$

Differential gain A_d

Apply $v_{in1} = +v_d/2$ and $v_{in2} = -v_d/2$. Whatever extra current Q_1 draws, Q_2 gives up exactly the same, so the current in R_E does not change — the emitter node behaves as an **AC ground**. Each half is then a simple common-emitter stage with emitter resistance r_e:

$i_{e1} = (v_d/2)/r_e$, $v_{o1} = -R_C i_{e1} = -R_C v_d/(2r_e)$, $v_{o2} = +R_C v_d/(2r_e)$

FORMULA — DIFFERENTIAL MODE

$$\text{Balanced: } A_d = \frac{v_{o2} - v_{o1}}{v_d} = \frac{R_C}{r_e} \qquad \text{Unbalanced: } A_d = \frac{R_C}{2r_e} \qquad R_i = 2\beta r_e, \quad r_e = \frac{V_T}{I_E}$$

Input resistance: the differential signal source sees base 1 → r_{e1} → r_{e2} → base 2. Each r_e looks β times bigger from the base, so $R_i = \beta(r_e + r_e) = 2\beta r_e$.

Common-mode gain A_{cm} and CMRR

Now apply the same v_{cm} to both bases. Both emitter currents rise together, so R_E carries $2i_e$. To each half it looks like a resistance $2R_E$ (twice the current through R_E = same drop as i_e through $2R_E$):

$i_e = v_{cm}/(r_e + 2R_E)$, $v_{o1} = v_{o2} = -R_C v_{cm}/(r_e + 2R_E)$

With a **balanced** output the two collectors move together, so $v_o = 0$ and $A_{cm} = 0$ in a perfectly matched circuit. With an **unbalanced** output:

FORMULA — COMMON MODE & CMRR (UNBALANCED OUTPUT)

$$A_{cm} = \frac{R_C}{r_e + 2R_E} \approx \frac{R_C}{2R_E} \qquad \text{CMRR} = \frac{A_d}{A_{cm}} = \frac{r_e + 2R_E}{2r_e} \approx \frac{R_E}{r_e}$$

$$\text{CMRR (dB)} = 20 \log_{10}(A_d/A_{cm})$$

KEY IDEA

CMRR ≈ R_E/r_e: **bigger tail resistance = better rejection**. But a huge R_E needs a huge $-V_{EE}$ to keep the same current. The way out: a **constant current source**, which has a very large AC resistance but only needs a few volts across it.

SOLVED EXAMPLE 1.2 · AC ANALYSIS (SAME CIRCUIT)

For the amplifier of Example 1.1 find r_e, A_d (balanced), R_i; and, if the output is taken from one collector, A_d, A_{cm} and CMRR in dB. Take V_T = 26 mV.

- r_e = 26 mV / 0.689 mA = 37.7 Ω
- Balanced: $A_d = R_C/r_e = 5600/37.7 = 148.4$; $R_i = 2\beta r_e = 2 \times 150 \times 37.7 = 11.3$ kΩ
- Unbalanced: $A_d = 148.4/2 = 74.2$
- $A_{cm} = 5600/(37.7 + 16\,400) = 0.341$

- CMRR = 74.2/0.341 = 217.8 ⇒ 20 log(217.8) = 46.8 dB

A_d = 148.4 (balanced) / 74.2 (unbalanced), R_i ≈ 11.3 kΩ, A_{cm} ≈ 0.34, CMRR ≈ 46.8 dB

SOLVED EXAMPLE 1.3 · OUTPUT WITH CMRR

A difference amplifier has A_d = 2000 and CMRR = 60 dB. Find the output when v_1 = 150 µV and v_2 = 140 µV. How much of it is error?

- CMRR = $10^{60/20}$ = 1000 ⇒ A_{cm} = 2000/1000 = 2
- v_d = 10 µV, v_{cm} = (150 + 140)/2 = 145 µV
- v_o = 2000 × 10 µV + 2 × 145 µV = 20 mV + 0.29 mV = 20.29 mV
- Error = 0.29/20 = 1.45 % — purely due to finite CMRR

v_o = 20.29 mV (0.29 mV is the common-mode error)

COMMON MISTAKE

Using $v_{cm} = v_1 + v_2$. The common-mode voltage is the **average**, not the sum. Also remember CMRR in dB must be converted (÷20, not ÷10) before dividing.

1.4 Constant current source

A current source is a circuit whose current stays the same whatever voltage appears across it. Seen by a small signal it looks like an **enormous resistance** (ideally infinite), which is exactly what the tail of a diff amp wants.

In Fig 1.2, R_1 and R_2 fix the base of Q_3. Its emitter sits one V_{BE} lower, so the voltage across R_E — and therefore the current — is fixed. The collector can swing up and down without changing I_T.

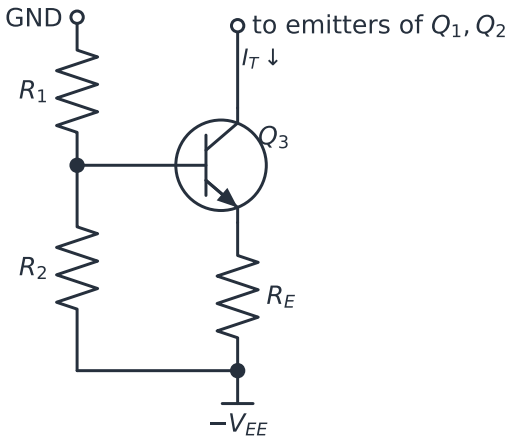


Fig 1.2 Constant current source for the tail.

FORMULA

$$V_{B3} = -\frac{R_1}{R_1 + R_2} V_{EE}$$

$$I_T = \frac{V_{E3} + V_{EE}}{R_E} = \frac{V_{EE} R_2/(R_1 + R_2) - V_{BE}}{R_E}$$

Because V_{BE} falls by about 2 mV/°C, designers often put a diode (or two) in series with R_1 — the diode drop changes the same way and cancels the drift. A Zener in place of R_2 does the same job and also fixes the current against supply variations.

SOLVED EXAMPLE 1.4

In Fig 1.2, V_{EE} = 15 V, $R_1 = R_2$ = 4.7 kΩ, V_{BE} = 0.7 V. Choose R_E for I_T = 1 mA (neglect base current).

- $V_{B3} = -15 \times 4.7/9.4 = -7.5$ V ⇒ $V_{E3} = -8.2$ V
- Voltage across $R_E = -8.2 - (-15) = 6.8$ V ⇒ $R_E = 6.8$ V / 1 mA

R_E = 6.8 kΩ (each diff-pair transistor then carries 0.5 mA)

1.5 Current mirrors — basic, Wilson, Widlar

Inside an IC, resistors are expensive (they eat silicon area) but transistors are cheap and beautifully matched. A **current mirror** copies a reference current from one branch into another using matched transistors.

THINK OF IT LIKE

A photocopier for current. You set the original (I_{ref}) with one resistor; every matched transistor whose base-emitter junction sees the same V_{BE} prints an identical copy.

Basic current mirror

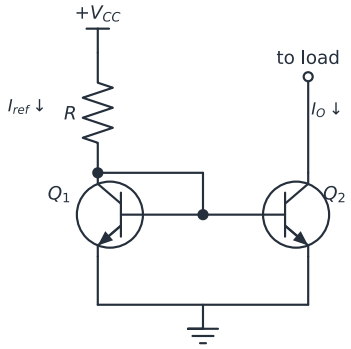


Fig 1.3 Basic current mirror; Q_1 is diode-connected.

Q_1 has its collector shorted to its base, so it behaves like a diode and develops exactly the V_{BE} needed to carry I_{ref}. Q_2 shares that V_{BE}, so (being identical) it carries the same collector current.

KCL at Q_1's collector: $I_{ref} = I_C + I_{B1} + I_{B2} = I_C + 2I_C/\beta$

FORMULA — BASIC MIRROR

$$I_{ref} = \frac{V_{CC} - V_{BE}}{R} \qquad I_O = \frac{I_{ref}}{1 + 2/\beta} \approx I_{ref} \qquad R_o = r_{o2}$$

Two weaknesses: (1) the small $2/\beta$ error, and (2) the output resistance is only r_o of Q_2 (Early effect), so I_O changes a little with output voltage.

SOLVED EXAMPLE 1.5

Design a basic mirror to give I_{ref} = 1 mA from V_{CC} = 15 V (V_{BE} = 0.7 V). Find I_O if β = 100.

- R = (15 − 0.7)/1 mA = 14.3 kΩ
- I_O = 1 mA/(1 + 2/100) = 1/1.02 = 0.980 mA (2 % low)

R = 14.3 kΩ, I_O ≈ 0.980 mA

Wilson current mirror

The Wilson mirror adds a third transistor Q_3 and uses negative feedback to fix both weaknesses. If I_O tries to rise, the voltage at Q_2's collector (diode-connected) rises, Q_1 sinks more current, the base of Q_3 is pulled down, and I_O comes back.

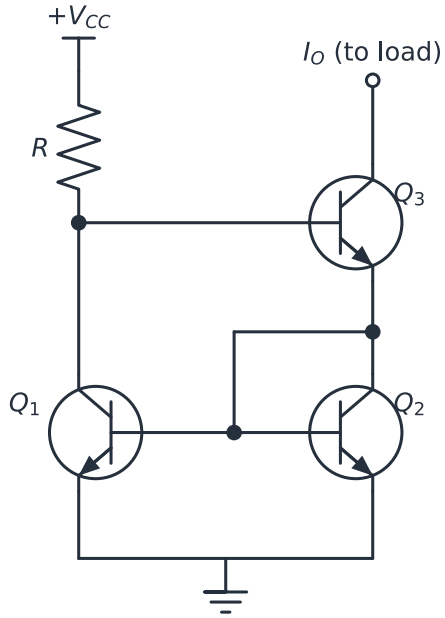


Fig 1.4 Wilson current mirror.

$I_{C1} = I_{C2} = I_C$ (bases tied). $I_{E3} = I_C + 2I_C/\beta \Rightarrow I_O = I_{E3}\cdot\beta/(\beta + 1) = I_C(\beta + 2)/(\beta + 1)$

$I_{ref} = I_{C1} + I_{B3} = I_C + I_O/\beta$. Dividing and simplifying:

FORMULA — WILSON MIRROR

$$I_O = I_{ref}\frac{\beta(\beta + 2)}{\beta^2 + 2\beta + 2} = I_{ref}\left(1 - \frac{2}{\beta^2 + 2\beta + 2}\right)$$

$$R_o \approx \beta r_o/2$$

For β = 100 the error drops from 2 % to just 0.02 %, and the output resistance goes up by roughly β/2.

Widlar current source

Suppose you need a tiny current like 10 µA. With a basic mirror you would need R ≈ 1.4 MΩ — far too large for an IC. Widlar's trick: add a small resistor R_E in Q_2's emitter so Q_2 gets a slightly smaller V_{BE}, and since current is exponential in V_{BE}, a few tens of millivolts reduce the current enormously.

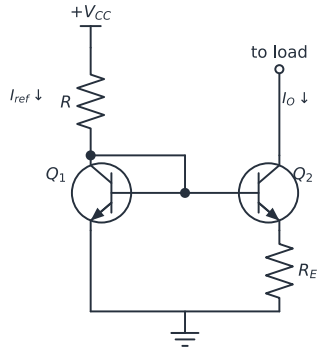


Fig 1.5 Widlar current source.

KVL in the base loop: $V_{BE1} = V_{BE2} + I_O R_E$
With $V_{BE} = V_T \ln(I_C/I_S)$ for matched transistors, $V_{BE1} - V_{BE2} = V_T \ln(I_{ref}/I_O)$:

FORMULA — WIDLAR

$$I_O R_E = V_T \ln \frac{I_{ref}}{I_O}$$

$$R_o \approx r_o(1 + g_m R_E)$$

The emitter resistor also raises output resistance (emitter degeneration), so the Widlar source is a better current source too.

SOLVED EXAMPLE 1.6 · GATE-STYLE

A Widlar source has I_{ref} = 1 mA. Find R_E for I_O = 10 µA (V_T = 26 mV). Compare with the resistor a basic mirror would need from a 15 V supply.

- $R_E = (V_T/I_O) \ln(I_{ref}/I_O) = (26\text{ mV}/10\ \mu\text{A}) \times \ln 100 = 2600 \times 4.605$
- R_E = 11.97 kΩ. The reference resistor is still 14.3 kΩ (for 1 mA).
- Basic mirror for 10 µA directly: R = 14.3 V/10 µA = 1.43 MΩ

R_E ≈ 12 kΩ — total ≈ 26 kΩ instead of 1.43 MΩ

EXAM TIP

Widlar equation is transcendental when I_O is unknown (given R_E). Solve by trial: guess I_O, compute the right side, iterate 2–3 times. Examiners accept this.

1.6 Active load and level shifter

Active load

Gain of a diff pair is $g_m R_C$. To get a big gain we want a big R_C — but a big resistor drops a big DC voltage and wastes area. Replace R_C with a **current mirror made of PNP transistors** (Fig 1.6). DC-wise it supplies the bias current with only ~0.7 V across it; AC-wise it looks like the transistor output resistance r_o (tens of kΩ to MΩ).

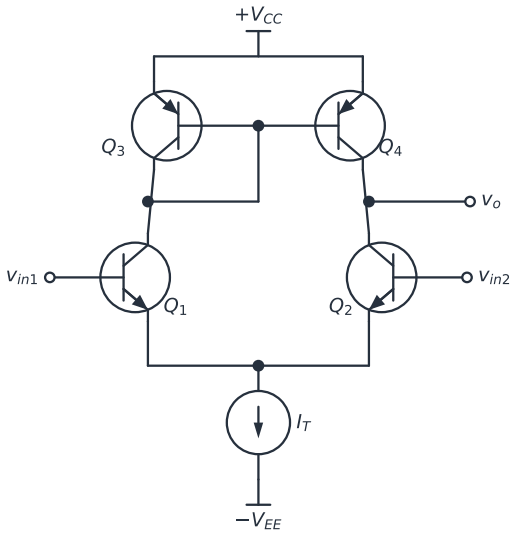


Fig 1.6 Diff pair with PNP current-mirror (active) load.

Bonus: the mirror copies Q_1's signal current over to the output node, so the single-ended output gets the *full* differential gain instead of half:

FORMULA

$$A_d = g_m(r_{o2} \parallel r_{o4}), \quad g_m = I_C/V_T$$

With $r_o \approx 100\ k\Omega$ and $I_C = 0.5\ mA$, $A_d \approx 19\ mS \times 50\ k\Omega \approx 960$ — from a single stage.

Level shifter

Op-amp stages are **direct-coupled** (no coupling capacitors — they would block DC and are too big for ICs). Each gain stage lifts the DC level, so after two stages the signal may be riding on several volts. If this reached the output, v_o would not be zero for zero input. The **level shifter** brings the DC level back to 0 V without reducing the AC signal.

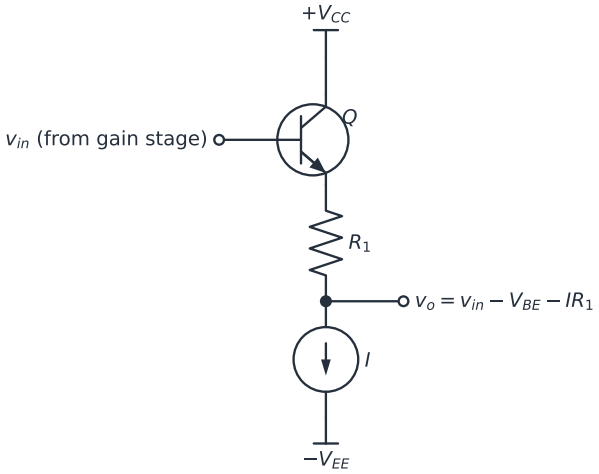


Fig 1.7 Emitter follower + current source as level shifter.

FORMULA

$$V_{o(DC)} = V_{in(DC)} - V_{BE} - IR_1$$

$$\text{AC gain} \approx 1$$

The current source has a very high r_o, so almost no AC signal is lost across R_1. Example: a 5 V DC level with $V_{BE} = 0.7\ V$ and $I = 1\ mA$ needs $IR_1 = 4.3\ V$, i.e. $R_1 = 4.3\ k\Omega$, to land at exactly 0 V.

INTERVIEW CORNER

Q. Why are current mirrors preferred over resistors for biasing in ICs?
A. Matched transistors are cheap in area and track each other with temperature; large resistors take huge area and have poor absolute tolerance (±20 %), while ratios of matched devices are accurate.

Q. Why does a Wilson mirror have higher output resistance?
A. It uses local negative (series-current) feedback: any change in I_O is sensed through Q_2 and corrected via Q_1 and Q_3, multiplying r_o by about $\beta/2$.

Q. What is an active load and why use it?
A. A transistor current source used as the collector load. It gives a very large AC resistance (high gain) with small DC drop and little area.

1.7 Block diagram of an op-amp

An **operational amplifier** is a direct-coupled, very-high-gain differential amplifier. The name comes from analog computers, where it performed mathematical *operations* — adding, integrating, differentiating. Every general-purpose op-amp is built from four cascaded stages:

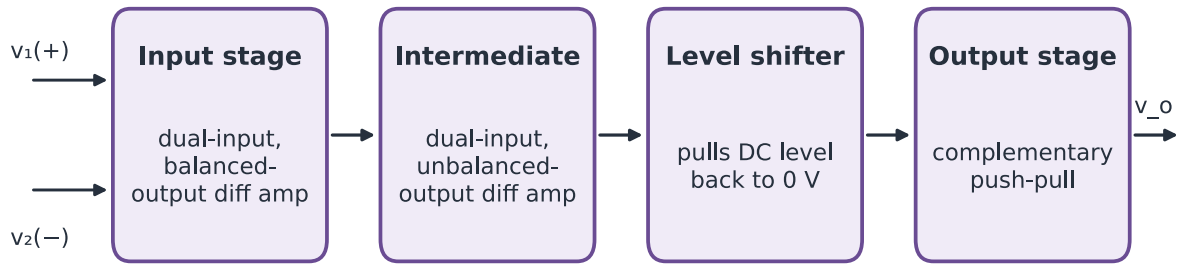


Fig 1.8 The four stages inside an op-amp.

Stage	Circuit	Job
Input stage	Dual-input balanced-output diff amp	Most of the voltage gain; sets input resistance, CMRR, offset and bias current
Intermediate stage	Dual-input unbalanced-output diff amp (driven by the first)	Extra gain; converts to single-ended
Level shifter	Emitter follower + current source	Shifts DC level back to 0 V
Output stage	Complementary push-pull (class AB)	Low output resistance, large swing, current to the load, short-circuit protection

The IC 741 (Fairchild, 1968) implements this with about 20 transistors, 11 resistors and a single 30 pF compensation capacitor. It runs from ±15 V (works from ±5 V to ±18 V) and comes in an 8-pin package.

Pin	1	2	3	4	5	6	7	8
Function	Offset null	Inverting (−)	Non-inverting (+)	$-V_{EE}$	Offset null	Output	$+V_{CC}$	NC

1.8 IC 741 — ideal vs practical parameters

An **ideal** op-amp is a mathematician's dream: infinite gain, infinite input resistance, zero output resistance, infinite bandwidth. Real chips come close enough that the ideal model gives answers within a percent in most circuits — but the differences matter for precision work and for interviews.

Parameter	Ideal	IC 741 (typical)	What it means
Open-loop gain A	∞	2×10^5 (106 dB)	gain with no feedback
Input resistance R_i	∞	2 MΩ	resistance between the inputs
Output resistance R_o	0	75 Ω	Thevenin resistance at output
Bandwidth	∞	$f_o \approx 5$ Hz open loop; $f_t = 1$ MHz	GBW product = 1 MHz
CMRR	∞	90 dB	rejection of common signals
PSRR (SVRR)	0 µV/V	30 µV/V (max 150)	sensitivity to supply changes
Slew rate	∞	0.5 V/µs	fastest output change
Input offset voltage V_{io}	0	2 mV (max 6 mV)	DC error referred to input
Input bias current I_B	0	80 nA (max 500 nA)	average base current
Input offset current I_{io}	0	20 nA (max 200 nA)	mismatch of base currents
Output swing (±15 V)	±15 V	±13 V	saturation levels
Drift	0	few µV/°C	change of offsets with temperature

1.9 DC errors: offset voltage, bias and offset current

Ground both inputs of a real op-amp and the output still drifts to $+V_{sat}$ or $-V_{sat}$. Why? The two input transistors are never perfectly identical, and they need small base currents. These cause DC errors.

- **Input offset voltage V_{io}** — the small DC voltage you must apply between the inputs to make the output exactly zero. It comes from V_{BE} mismatch of the input pair.
- **Input bias current I_B** = $(I_{B1} + I_{B2})/2$ — the average of the two base currents when the output is zero.
- **Input offset current I_{io}** = $|I_{B1} - I_{B2}|$ — the mismatch of the two base currents.

Output offset voltage in an inverting / non-inverting amplifier

Due to V_{io}: V_{io} behaves like a tiny source in series with the + input, and the circuit amplifies it by the non-inverting gain: $V_{oo} = (1 + R_f/R_1)V_{io}$.

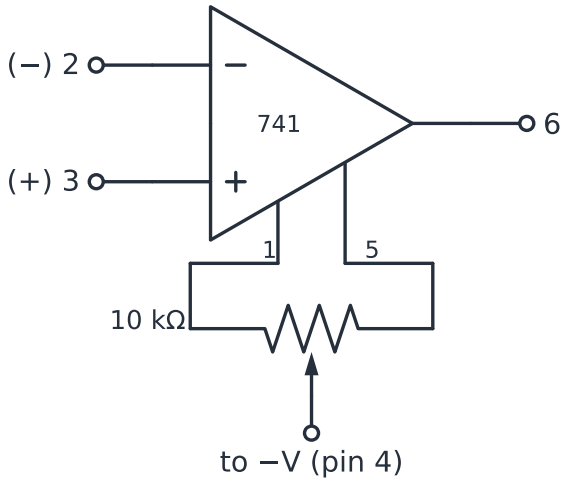
Due to I_B: with + input grounded, $V_- \approx 0$, so no current flows in R_1; I_{B1} must come through R_f from the output: $V_{oo} = I_{B1}R_f$.

Cure — bias-current compensation: put R_{comp} in the + lead (Fig 1.10). I_{B2} drops $-I_{B2}R_{comp}$ at the + input. Working through KCL, the output becomes $V_{oo} = R_fI_{B1} - I_{B2}R_{comp}(1 + R_f/R_1)$. Choosing $R_{comp} = R_1R_f/(R_1 + R_f)$ makes the second term exactly R_fI_{B2}, leaving only the *mismatch*:

FORMULA — OUTPUT OFFSET

$$R_{comp} = R_1 \parallel R_f \;\Rightarrow\; V_{oo}(\text{bias}) = R_fI_{io}$$

$$\text{Total worst case: } V_{oo} = \left(1 + \frac{R_f}{R_1}\right)V_{io} + R_fI_{io}$$



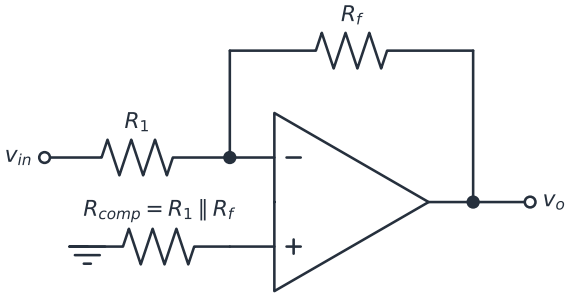


Fig 1.10 Bias-current compensation.

Fig 1.9 Offset nulling of the 741 with a 10 kΩ pot.

V_{io} itself is nulled with a 10 kΩ potentiometer between pins 1 and 5, wiper to $-V_{EE}$ (Fig 1.9): it slightly unbalances the input-stage emitter resistors until the output reads zero.

SOLVED EXAMPLE 1.7 · OUTPUT OFFSET

An inverting amplifier uses R_1 = 10 kΩ, R_f = 100 kΩ and a 741 with V_{io} = 6 mV, I_B = 500 nA, I_{io} = 200 nA (max values). Find the worst-case output offset (a) without and (b) with bias-current compensation. Give R_{comp}.

- From V_{io}: (1 + 100/10) × 6 mV = 11 × 6 = 66 mV
- (a) From bias current: I_BR_f = 500 nA × 100 kΩ = 50 mV ⇒ total = 116 mV
- (b) R_{comp} = 10 k ∥ 100 k = 9.09 kΩ; bias part becomes $I_{io}R_f$ = 200 nA × 100 k = 20 mV ⇒ total = 86 mV

(a) 116 mV (b) 86 mV, $R_{comp} \approx 9.1\ k\Omega$ (null the remaining 66 mV with the pot)

1.10 Slew rate, GBW, PSRR and frequency compensation

Slew rate

Slew rate (SR) is the maximum rate at which the output can change: $SR = (dv_o/dt)_{max}$, in V/µs. Inside the 741, the internal stage can only push a limited current I (≈ 15 µA) into the 30 pF compensation capacitor, so $dv/dt \le I/C = 0.5$ V/µs. Ask for a faster change and the output turns into a triangle (Fig 1.11).

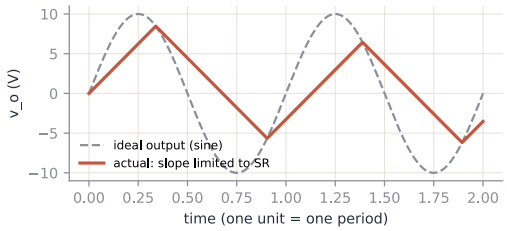


Fig 1.11 Slew-rate limiting: a large, fast sine becomes a triangle.

For $v_o = V_m \sin \omega t$ the steepest slope is ωV_m (at the zero crossing). No distortion needs $\omega V_m \le SR$:

FORMULA — SLEW RATE

$$SR = \frac{I_{max}}{C_c} \qquad f_{max} = \frac{SR}{2\pi V_m} \quad \text{(full-power bandwidth)}$$

SOLVED EXAMPLE 1.8 · SLEW RATE

A 741 (SR = 0.5 V/µs) must deliver a 10 V peak sine. (a) Highest undistorted frequency? (b) At 40 kHz, what is the largest undistorted peak?

- (a) $f_{max} = 0.5 \times 10^6/(2\pi \times 10) = 7.96$ kHz
- (b) $V_m = SR/(2\pi f) = 0.5 \times 10^6/(2\pi \times 40 \times 10^3) = 1.99$ V

(a) ≈ 7.96 kHz (b) ≈ 1.99 V peak

Gain-bandwidth product (GBW)

Because of the compensation capacitor, the 741's open-loop gain is 2×10^5 only up to about 5 Hz; then it falls at −20 dB/decade and reaches 1 (0 dB) at $f_t = 1$ MHz. On this slope, **gain × frequency = constant = f_t**. So with feedback, if you settle for a closed-loop gain A_{CL}, the bandwidth stretches to f_t/A_{CL}.

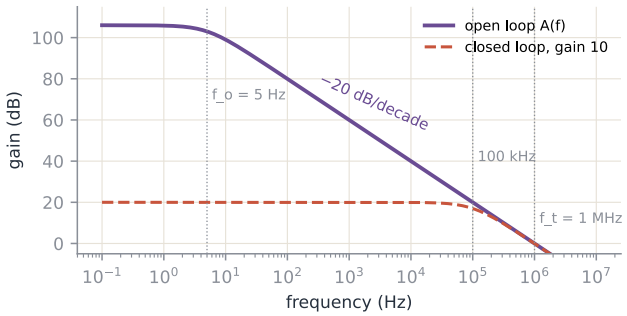


Fig 1.12 741 open-loop response and a closed-loop gain of 10 (20 dB) with 100 kHz bandwidth.

FORMULA — GBW

$$A_o \times f_o = A_{CL} \times f_{CL} = f_t \quad (741{:}\ 2 \times 10^5 \times 5\ \text{Hz} = 1\ \text{MHz})$$

SOLVED EXAMPLE 1.9

A 741 is used as an amplifier with gain 50. What is its bandwidth? If the supply ripple is 2 V and PSRR = 150 µV/V, what output ripple appears?

- $f_{CL} = 1\ \text{MHz}/50 = 20\ \text{kHz}$
- Input-referred ripple = 150 µV/V × 2 V = 300 µV; it is amplified by the noise gain 50 ⇒ 15 mV at the output

BW = 20 kHz; output ripple ≈ 15 mV

CMRR and PSRR

CMRR = A_d/A_{cm} (90 dB for the 741 — a common signal is attenuated about 31 600 times relative to a difference signal). **PSRR** (power-supply rejection ratio, also called SVRR) = $\Delta V_{io}/\Delta V_{supply}$, the change in input offset per volt of supply change. Smaller µV/V is better — the opposite sense of CMRR, so be careful with wording.

Frequency compensation — the basics

An op-amp has several internal poles (each stage adds one), and each pole adds up to −90° of phase. With two or three poles the total phase can hit −180° while the loop gain is still above 1. Negative feedback then turns into positive feedback and the amplifier **oscillates**.

Compensation deliberately shapes the response so that the loop gain falls below unity before the phase reaches −180°. The usual measure is **phase margin** = 180° + phase at the frequency where loop gain = 1; ≥ 45° is considered safe.

- **Dominant-pole compensation** (741): a capacitor creates one very low-frequency pole (≈ 5 Hz) so the gain is down to 1 before the second pole arrives. The 30 pF is placed across the second stage, where the **Miller effect** multiplies it into a large effective capacitance.

- **Pole-zero (lag) and lead compensation**: add an RC network (externally on chips like LM301 or 709) to cancel a pole or add phase lead — gives more bandwidth than a brute-force dominant pole.
- **Internally compensated** (741): stable down to gain 1, but slow. **Externally/decompensated** op-amps are faster when used at high gain.

INTERVIEW CORNER

Q. Why does the 741 have such a low open-loop bandwidth (≈ 5 Hz)?
A. Dominant-pole compensation by the internal 30 pF capacitor, which guarantees stability for any resistive feedback including unity gain.

Q. Slew rate vs bandwidth — which limits a large-signal sine?
A. For large outputs, slew rate (full-power bandwidth $SR/2\pi V_m$) usually limits first; for small signals, GBW limits.

Q. What causes input offset voltage?
A. Mismatch in V_{BE} (and collector resistors) of the input differential pair.

Q. If the open-loop gain is infinite, how can the output be finite?
A. With negative feedback $v_d = v_o/A \to 0$: the inputs are forced to (almost) the same voltage and the output takes whatever value the feedback network demands.

1.11 Equivalent circuit and voltage transfer curve

For circuit analysis we model the op-amp as a black box (Fig 1.13): an input resistance R_i between the inputs, and on the output side a voltage-controlled voltage source $A \cdot v_d$ behind an output resistance R_o. The ideal op-amp is this model with $R_i \to \infty$, $R_o \to 0$ and $A \to \infty$.

The **voltage transfer curve** (Fig 1.14) plots v_o against v_d: a straight line of slope A through the origin that clips at $\pm V_{sat}$ (about 2 V inside the rails). With $A = 2 \times 10^5$ and $V_{sat} = 13$ V, the linear region is only **$\pm 13/(2 \times 10^5) = \pm 65$ µV** wide — smaller than the noise. So in open loop the op-amp is practically a **comparator**; to use it as an amplifier we **must** add negative feedback (Module 2).

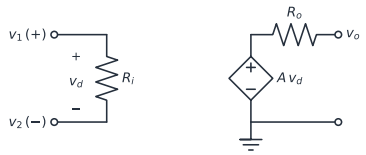


Fig 1.13 Equivalent circuit: $v_o = A(v_1 - v_2)$ unloaded.

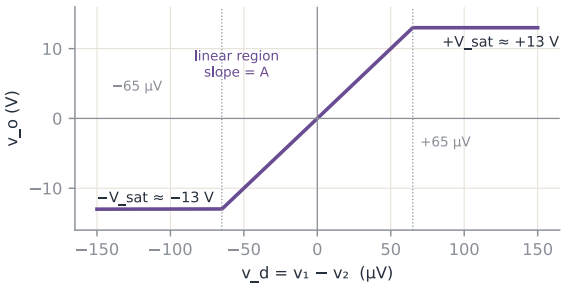


Fig 1.14 Voltage transfer curve of a 741 on ±15 V.

✓ Module 1 — Quick revision

DC: $I_E = (V_{EE} - V_{BE})/(2R_E + R_{in}/\beta)$; $V_{CE} = V_{CC} + V_{BE} - I_C R_C$

r_e = 26 mV/I_E. **A_d:** R_C/r_e balanced; $R_C/2r_e$ unbalanced. $R_i = 2\beta r_e$

A_{cm} (unbalanced) = $R_C/(r_e + 2R_E)$; balanced ideally 0

CMRR ≈ R_E/r_e; dB = $20 \log(A_d/A_{cm})$. $v_o = A_d v_d + A_{cm} v_{cm}$

Current source: $I_T = (V_{EE} R_2/(R_1+R_2) - V_{BE})/R_E$
Basic mirror: $I_O = I_{ref}/(1 + 2/\beta)$; $I_{ref} = (V_{CC} - V_{BE})/R$

Wilson: $I_O = I_{ref}(1 - 2/(\beta^2+2\beta+2))$; $R_o \approx \beta r_o/2$
Widlar: $I_O R_E = V_T \ln(I_{ref}/I_O)$ — small currents with small R

Active load: $A_d = g_m(r_{o2} \parallel r_{o4})$. **Level shifter:** $V_o = V_{in} - V_{BE} - IR_1$

Op-amp stages: diff amp (balanced) → diff amp (unbalanced) → level shifter → push-pull

741: $A = 2\times10^5$, R_i = 2 MΩ, R_o = 75 Ω, SR = 0.5 V/µs, f_t = 1 MHz, CMRR 90 dB, V_{io} 2 mV, I_B 80 nA, I_{io} 20 nA

Offset: $V_{oo} = (1+R_f/R_1)V_{io} + R_f I_{io}$ with $R_{comp} = R_1 \| R_f$

Slew: SR = I/C_c; $f_{max} = SR/(2\pi V_m)$. **GBW:** $A_{CL} f_{CL} = f_t$

VTC: linear only for $|v_d| < V_{sat}/A \approx 65$ µV.
Compensation: dominant pole ⇒ phase margin ≥ 45°

Practice — short answer (3 marks)

1. Define CMRR. Why is it infinite for an ideal dual-input balanced-output diff amp?
2. Why is a constant current source preferred over R_E in a diff amp?
3. Draw a basic current mirror and show that $I_O = I_{ref}/(1 + 2/\beta)$.
4. What is the need for a level shifter in an op-amp?
5. Define input offset voltage, input bias current and input offset current.
6. Define slew rate. Find f_{max} for a 5 V peak output with SR = 0.5 V/µs. [15.9 kHz]
7. List five ideal op-amp characteristics with 741 values. Sketch the VTC.
8. Draw the block diagram of an op-amp and name the job of each stage.

Practice — long answer (9 / 14 marks)

1. DC and AC analysis of a dual-input balanced-output diff amp; derive A_d, R_i, R_o.
2. Derive A_{cm} and CMRR for a dual-input unbalanced-output diff amp. For R_C = 4.7 kΩ, R_E = 6.8 kΩ, $V_{CC} = V_{EE}$ = 10 V, β = 100, find all of them. [$I_E \approx 0.684$ mA, $r_e \approx 38$ Ω, $A_d \approx 61.8$, CMRR ≈ 45 dB]
3. Derive the output current of a Wilson mirror; compare with the basic mirror.
4. Explain the Widlar current source. Design it for I_O = 20 µA with I_{ref} = 2 mA. [$R_E \approx 5.99$ kΩ]
5. Effect of V_{io}, I_B on an inverting amp; derive R_{comp}. Explain frequency compensation.

02

Negative Feedback & Op-amp Applications

> *This is the module that turns a ±65 µV comparator into precise amplifiers, integrators, rectifiers and converters — and it supplies half the questions in every analog interview.*

SYLLABUS CHECKLIST

- ☐ Inverting and non-inverting amplifiers, voltage follower
- ☐ Effect of negative feedback on gain, input/output impedance and bandwidth
- ☐ Summing and difference amplifiers; instrumentation amplifier
- ☐ Integrator and differentiator — ideal and practical
- ☐ V-to-I and I-to-V converters
- ☐ Precision half-wave and full-wave rectifiers
- ☐ Comparators; Schmitt trigger (inverting / non-inverting, hysteresis)
- ☐ Log and antilog amplifiers; peak detector; sample and hold

≈ 9 hours · CO2 · Design linear and non-linear op-amp application circuits

2.1 Inverting, non-inverting amplifiers and voltage follower

Connect part of the output back to the **inverting** input and something magical happens: the op-amp keeps adjusting its output until the two inputs are equal. That single behaviour gives the two golden rules used in every circuit of this module:

KEY IDEA — THE TWO GOLDEN RULES (NEGATIVE FEEDBACK, IDEAL OP-AMP)

Rule 1 (virtual short): $v_+ = v_-$, because $v_d = v_o/A \to 0$. **Rule 2:** no current enters either input, because $R_i \to \infty$.

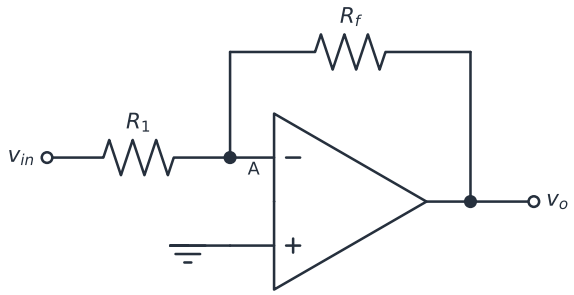


Fig 2.1 Inverting amplifier (node A is a virtual ground).

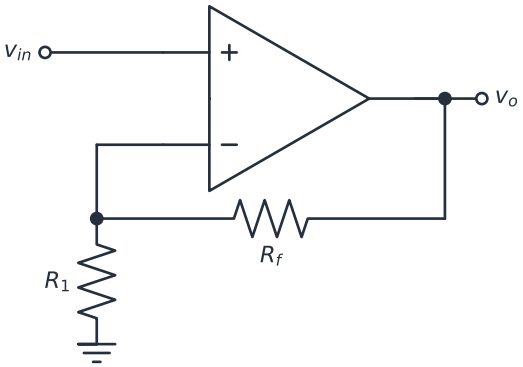


Fig 2.2 Non-inverting amplifier.

Inverting (Fig 2.1). + is grounded, so by Rule 1 node A is at 0 V — a **virtual ground** (at 0 V but not connected to ground). The current in R_1 is v_{in}/R_1; by Rule 2 it all flows on through R_f, so $v_o = 0 - (v_{in}/R_1)R_f$.

Non-inverting (Fig 2.2). The input goes to +. R_f and R_1 divide the output: $v_- = v_o R_1/(R_1 + R_f)$. Rule 1 says this equals v_{in}.

FORMULA — THE TWO BASIC AMPLIFIERS

$$\text{Inverting: } A_v = -\frac{R_f}{R_1}, \quad R_{in} = R_1 \qquad \text{Non-inverting: } A_v = 1 + \frac{R_f}{R_1}, \quad R_{in} \to \infty$$

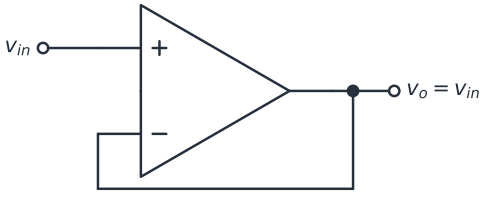


Fig 2.3 Voltage follower.

Voltage follower (Fig 2.3). Put $R_f = 0$ and $R_1 = \infty$ in the non-inverting amplifier: gain = 1, $v_o = v_{in}$. Why bother with a gain of one? Because it has almost infinite input resistance and almost zero output resistance — a perfect **buffer** between a weak source (a sensor, a capacitor holding a voltage) and a heavy load.

SOLVED EXAMPLE 2.1

With R_1 = 10 kΩ and R_f = 100 kΩ, find v_o for v_{in} = 0.5 V in (a) the inverting and (b) the non-inverting connection. What happens for v_{in} = 2 V on ±15 V supplies?

- (a) $A_v = -100/10 = -10 \Rightarrow v_o = -5$ V; input resistance = 10 kΩ
- (b) $A_v = 1 + 10 = 11 \Rightarrow v_o = 5.5$ V
- v_{in} = 2 V would need −20 V / +22 V — beyond $\pm V_{sat} \approx \pm 13$ V, so the output clips at ∓ 13 V

(a) −5 V (b) +5.5 V (2 V input: saturates at ≈ ±13 V)

2.2 Effect of negative feedback on gain, impedance and bandwidth

Feedback "trades" gain for almost everything else. Let β = fraction of output fed back ($\beta = R_1/(R_1 + R_f)$ for both basic amplifiers) and let 1 + Aβ be the **desensitivity factor** (loop gain + 1). The non-inverting amplifier is **voltage-series** feedback; the inverting amplifier is **voltage-shunt** feedback.

FORMULA — NON-INVERTING (VOLTAGE-SERIES) AMPLIFIER WITH REAL OP-AMP

$$A_F = \frac{A}{1 + A\beta} \approx \frac{1}{\beta} \quad R_{iF} = R_i(1 + A\beta) \quad R_{oF} = \frac{R_o}{1 + A\beta} \quad f_F = f_o(1 + A\beta)$$

$$\text{Inverting: } A_F = -\frac{AR_f}{R_1 + R_f + AR_1}, \quad R_{iF} \approx R_1, \quad R_{oF} = R_o/(1 + A\beta), \quad f_F = f_t\beta$$

- **Gain** falls by (1 + Aβ) but becomes set by resistors only — stable against temperature and device changes.
- **Input impedance** rises (series feedback) — or, for shunt feedback, becomes ≈ R_1.
- **Output impedance** falls by (1 + Aβ) — voltage sampling makes the output a stiff voltage source.
- **Bandwidth** rises by the same factor, keeping gain × bandwidth = f_t. Distortion and noise inside the loop also fall.

SOLVED EXAMPLE 2.2 · REAL 741 IN A GAIN-11 NON-INVERTING AMPLIFIER

$A = 2 \times 10^5$, R_i = 2 MΩ, R_o = 75 Ω, f_o = 5 Hz; R_1 = 10 kΩ, R_f = 100 kΩ. Find A_F, R_{iF}, R_{oF}, f_F.

- $\beta = 10/110 = 1/11 \Rightarrow 1 + A\beta = 1 + 2 \times 10^5/11 = 18\,183$
- $A_F = 2 \times 10^5/18\,183 = 10.9994$ (ideal 11)
- $R_{iF} = 2$ MΩ × 18 183 = 36.4 GΩ; $R_{oF} = 75/18\,183 = 4.12$ mΩ
- $f_F = 5 \times 18\,183 = 90.9$ kHz (= 1 MHz/11 ✓)

$A_F \approx 11$, $R_{iF} \approx 36\ G\Omega$, $R_{oF} \approx 4\ m\Omega$, $f_F \approx 91\ kHz$

INTERVIEW CORNER

Q. Virtual ground vs real ground?

A. The inverting node sits at 0 V only because feedback forces it; no current flows into it to ground. Short it to ground and the circuit's behaviour changes (the feedback current would be diverted).

Q. Why is a buffer needed if gain is 1?

A. Impedance transformation: very high input impedance so the source isn't loaded, very low output impedance so the load doesn't change the voltage.

Q. Bandwidth of an inverting amp of gain −1 with f_t = 1 MHz?

A. $\beta = 1/2$, so BW = $f_t\beta$ = 500 kHz — not 1 MHz. The "noise gain" $(1 + R_f/R_1)$ sets bandwidth, not the signal gain.

2.3 Summing and difference amplifiers

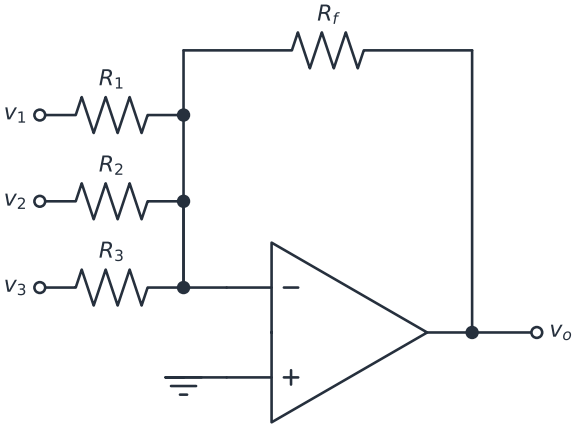


Fig 2.4 Inverting summing amplifier.

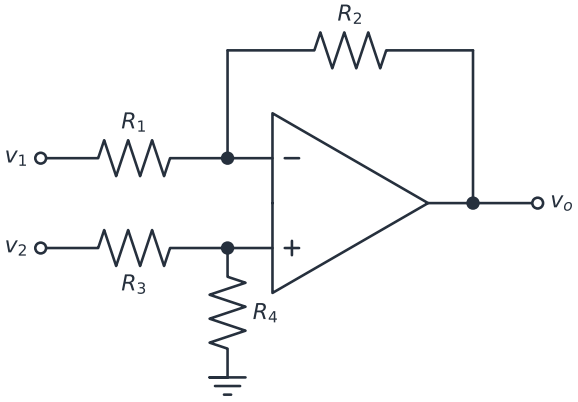


Fig 2.5 Difference (subtractor) amplifier.

Summer. The virtual ground isolates the inputs from each other: each input pushes its own current v_k/R_k into node A, and they all add up in R_f. Equal resistors give a plain adder; $R_f = R/n$ gives an **averager**; unequal R_k gives a **scaling (weighted) adder** — the idea behind the weighted-resistor DAC of Module 4.

Difference amplifier. Use superposition. With $v_2 = 0$ it's an inverting amp: $-(R_2/R_1)v_1$. With $v_1 = 0$, the + input sees $v_2R_4/(R_3 + R_4)$, amplified by $(1 + R_2/R_1)$.

FORMULA

$$\text{Summer: } v_o = -R_f\left(\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3}\right)$$

$$\text{Difference: } v_o = -\frac{R_2}{R_1}v_1 + \left(1 + \frac{R_2}{R_1}\right)\frac{R_4}{R_3 + R_4}v_2 \quad \rightarrow \quad v_o = \frac{R_2}{R_1}(v_2 - v_1) \text{ if } R_3 = R_1,\ R_4 = R_2$$

SOLVED EXAMPLE 2.3 · DESIGN A WEIGHTED SUMMER

Design a circuit giving $v_o = -(2v_1 + 3v_2 + 5v_3)$ with $R_f = 30\ k\Omega$. Find v_o for $v_1 = 1\ V$, $v_2 = -0.5\ V$, $v_3 = 0.2\ V$.

- $R_f/R_1 = 2 \Rightarrow R_1 = 15\ k\Omega$; $R_2 = 30/3 = 10\ k\Omega$; $R_3 = 30/5 = 6\ k\Omega$
- $v_o = -(2 - 1.5 + 1.0) = -1.5\ V$

$R_1 = 15\ k\Omega$, $R_2 = 10\ k\Omega$, $R_3 = 6\ k\Omega$; $v_o = -1.5\ V$

COMMON MISTAKE

The simple difference amplifier has **unequal and low input resistances** (R_1 at one input, $R_3 + R_4$ at the other), and its CMRR collapses if the resistor ratios differ by even 1 %. That is exactly why the instrumentation amplifier exists.

2.4 Instrumentation amplifier

Sensors such as strain-gauge bridges and ECG electrodes produce a few millivolts of *difference* riding on volts of common-mode voltage, through high source resistance. We need: very high input impedance on **both** inputs, very high CMRR, and gain set by **one** resistor. The three-op-amp instrumentation amplifier delivers all of it.

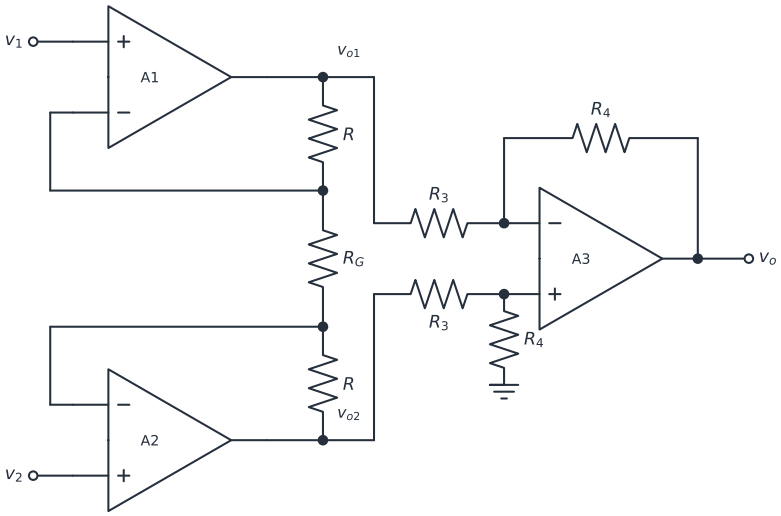


Fig 2.6 Three-op-amp instrumentation amplifier: buffered input stage (A1, A2) + difference amplifier (A3).

Derivation. By the virtual short, the − inputs of A1 and A2 sit at v_1 and v_2, so $v_1 - v_2$ appears across R_G. The current $i = (v_1 - v_2)/R_G$ cannot enter the op-amp inputs, so it flows through R, R_G, R in series:

$v_{o1} - v_{o2} = i(R + R_G + R) = (1 + 2R/R_G)(v_1 - v_2)$
A3 is a difference amp: $v_o = (R_4/R_3)(v_{o2} - v_{o1})$

FORMULA — INSTRUMENTATION AMPLIFIER

$$v_o = \frac{R_4}{R_3}\left(1 + \frac{2R}{R_G}\right)(v_2 - v_1)$$

A common-mode input moves both − inputs equally, so no current flows in R_G and the first stage passes it with gain 1; A3 then rejects it. The first stage amplifies the difference but not the common mode — CMRR is boosted by $(1 + 2R/R_G)$.

SOLVED EXAMPLE 2.4 · PLACEMENT FAVOURITE

In Fig 2.6, R = 25 kΩ and $R_3 = R_4 = 10$ kΩ. Find R_G for a gain of 101, and v_o when $v_1 = 2.000$ V, $v_2 = 2.015$ V.

- $1 + 2(25\text{ k})/R_G = 101 \Rightarrow R_G = 50\text{ k}/100 = 500\ \Omega$
- $v_o = 1 \times 101 \times (2.015 - 2.000) = 101 \times 15$ mV — the 2 V common-mode part is rejected

$R_G = 500\ \Omega$, $v_o = 1.515$ V

2.5 Integrator and differentiator

Swap a resistor for a capacitor and the op-amp starts doing calculus. A capacitor's current is $i = C\,dv/dt$, so whichever element carries the known current decides the operation.

Integrator

Ideal (no R_f): current v_{in}/R_1 charges C_f from the virtual ground, so $v_o = -(1/C_f)\int i\,dt$. A constant input gives a ramp; a square wave gives a triangle (Fig 2.9).

Problem: at DC the capacitor is open, so the gain is the open-loop gain. Any tiny offset or bias current is integrated until the output saturates. **Practical integrator:** add R_f across C_f (Fig 2.7) to limit the DC gain to $-R_f/R_1$. The circuit now integrates properly only above f_a.

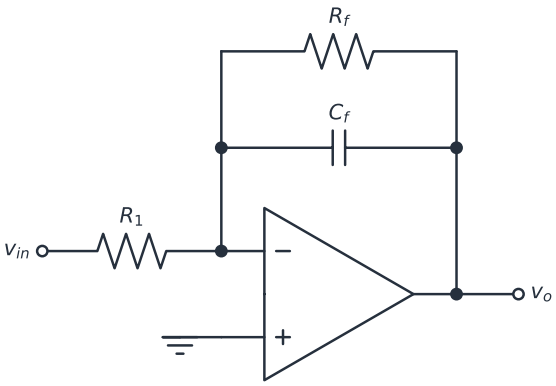


Fig 2.7 Practical integrator.

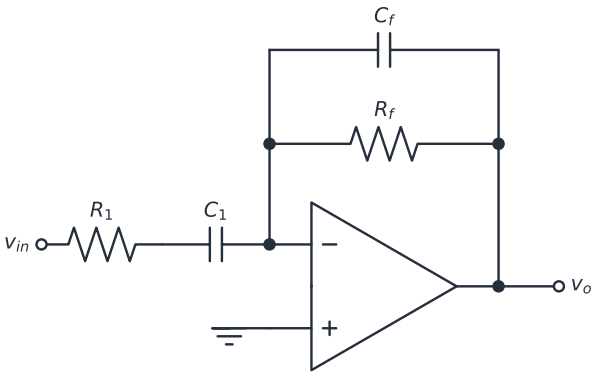


Fig 2.8 Practical differentiator.

FORMULA — INTEGRATOR

$$\text{Ideal: } v_o = -\frac{1}{R_1 C_f}\int v_{in}\,dt + v_o(0) \qquad |A| = \frac{1}{\omega R_1 C_f}\ (-20\ \text{dB/dec; 0 dB at } f_b = \frac{1}{2\pi R_1 C_f})$$

$$\text{Practical: } A(j\omega) = -\frac{R_f/R_1}{1 + j\omega R_f C_f}, \quad f_a = \frac{1}{2\pi R_f C_f}\text{; integrates well for } f \gg f_a \text{ (choose } R_f \approx 10R_1)$$

Differentiator

Ideal: C at the input, R_f in feedback: $v_o = -R_f C_1\, dv_{in}/dt$. Its gain rises at +20 dB/decade forever, so it amplifies high-frequency noise and tends to oscillate. **Practical**: add R_1 in series with C_1 and C_f across R_f (Fig 2.8) so the gain stops rising above f_b.

FORMULA — DIFFERENTIATOR DESIGN

$$\text{Ideal: } v_o = -R_f C_1 \frac{dv_{in}}{dt}, \quad 0\ \text{dB at } f_a = \frac{1}{2\pi R_f C_1}$$

$$\text{Practical: } f_b = \frac{1}{2\pi R_1 C_1}, \quad R_1 C_1 = R_f C_f\text{; choose } f_a = \text{highest input frequency, } f_b = 10\text{–}20\ f_a$$

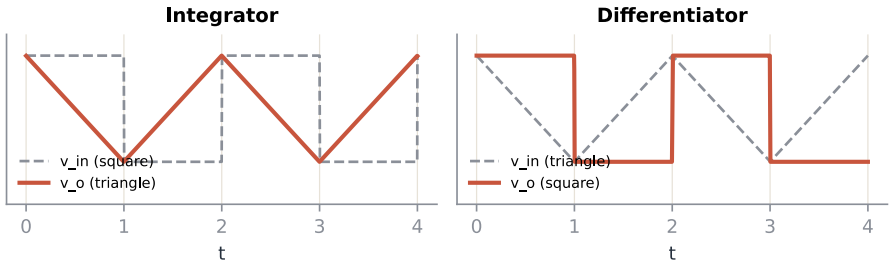


Fig 2.9 Square → triangle (integrator) and triangle → square (differentiator). Both invert.

SOLVED EXAMPLE 2.5 · INTEGRATOR

A practical integrator has R_1 = 10 kΩ, C_f = 0.1 µF, R_f = 100 kΩ. Find f_a and f_b. A ±1 V, 500 Hz square wave is applied: what is the peak-to-peak output triangle?

▸ $f_a = 1/(2\pi \times 100\text{ k} \times 0.1\ \mu) = 15.9$ Hz; $f_b = 1/(2\pi \times 10\text{ k} \times 0.1\ \mu) = 159$ Hz

- 500 Hz ≫ f_a, so it integrates. Each half-period T/2 = 1 ms the input is constant (1 V)
- $\Delta v_o = V \cdot (T/2)/(R_1 C_f) = 1 \times 1\text{ ms}/1\text{ ms} = 1\text{ V}$

$f_a \approx 15.9$ Hz, $f_b \approx 159$ Hz, output ≈ 1 V peak-to-peak triangle

SOLVED EXAMPLE 2.6 · DIFFERENTIATOR DESIGN

Design a practical differentiator for input signals up to 1 kHz using $C_1 = 0.1\ \mu$F. Take $f_b = 20\ f_a$.

- $f_a = 1$ kHz ⇒ $R_f = 1/(2\pi \times 1000 \times 0.1\ \mu) = 1.59$ kΩ
- $f_b = 20$ kHz ⇒ $R_1 = 1/(2\pi \times 20\,000 \times 0.1\ \mu) = 79.6\ \Omega$
- $C_f = R_1 C_1 / R_f = 79.6 \times 0.1\ \mu / 1592 = 5$ nF

$R_f \approx 1.6$ kΩ, $R_1 \approx 80\ \Omega$, $C_f = 5$ nF

2.6 V-to-I and I-to-V converters

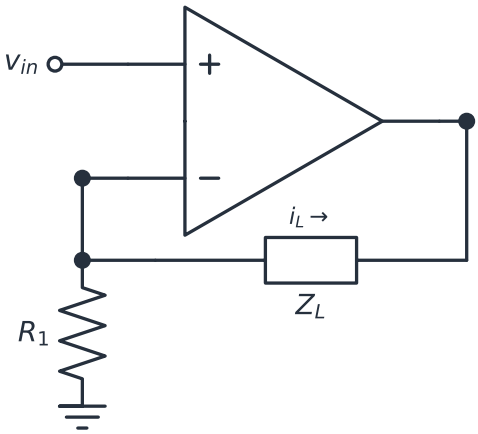


Fig 2.10 V-to-I converter, floating load.

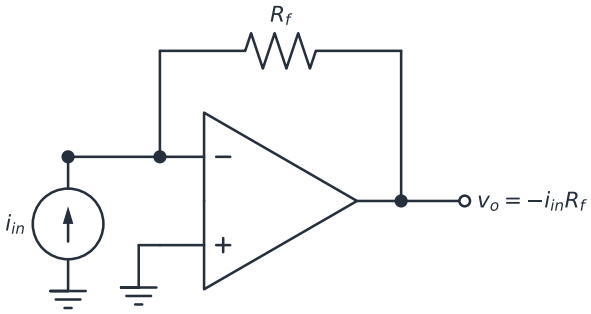


Fig 2.11 I-to-V (transimpedance) converter.

V-to-I (Fig 2.10). The − input follows v_{in}, so R_1 carries v_{in}/R_1. That current can only come through the load Z_L in the feedback path: **$i_L = v_{in}/R_1$, independent of Z_L**. Uses: LED/relay drivers, low-voltage DC voltmeters, diode testers. (For a grounded load the *Howland* current source is used.)

I-to-V (Fig 2.11). An input current flows into the virtual ground and through R_f: **$v_o = -i_{in}R_f$**. Because the input node is held at 0 V, the source (a photodiode, a current-output DAC) sees zero load resistance — ideal for current sources.

SOLVED EXAMPLE 2.7

In Fig 2.10, $v_{in} = 2$ V, $R_1 = 1$ kΩ, $V_{sat} = 13$ V. Find i_L and the largest load that can be driven.

- $i_L = 2/1\text{ k} = 2$ mA (any load)

▸ Output voltage = $v_{in} + i_L Z_L \le 13$ V ⇒ $Z_L \le (13 - 2)/2$ mA

i_L = 2 mA, $Z_{L(max)}$ = 5.5 kΩ

2.7 Precision half-wave and full-wave rectifiers

An ordinary diode cannot rectify a 0.2 V signal — it needs 0.7 V before it conducts. Put the diode **inside the feedback loop** and the op-amp's huge gain pushes its output to whatever is needed: the effective cut-in voltage becomes 0.7 V/A ≈ a few µV. This is a **precision rectifier** or "super diode".

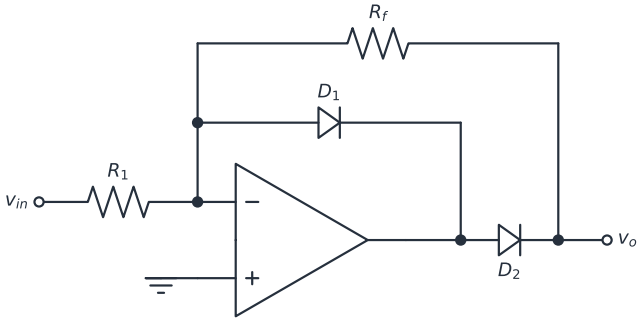


Fig 2.12 Precision (inverting) half-wave rectifier.

Working (Fig 2.12). For $v_{in} > 0$ the op-amp output swings negative; D_1 conducts and clamps the inverting node, D_2 is off, so $v_o = 0$. For $v_{in} < 0$ the output swings positive; D_2 conducts, D_1 is off, and the circuit is an inverting amplifier: $v_o = -(R_f/R_1)v_{in} > 0$. D_1 also stops the op-amp from saturating, so it recovers quickly.

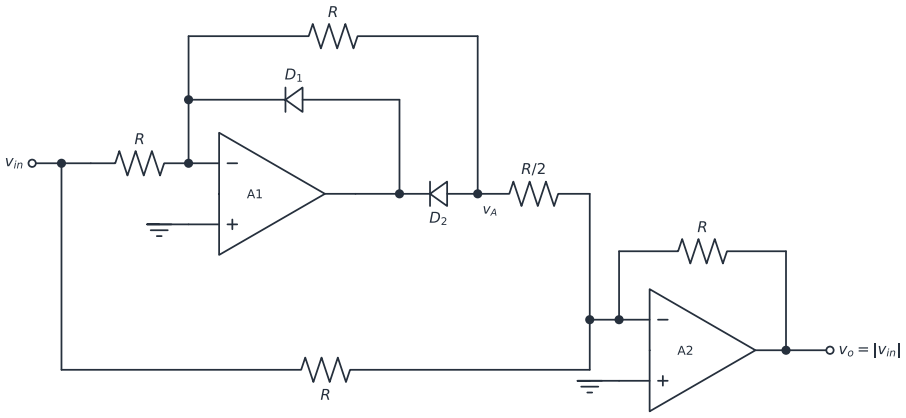


Fig 2.13 Precision full-wave rectifier (absolute-value circuit): HWR + summer.

Full-wave (Fig 2.13). Here the HWR's diodes are turned round so that $v_A = -v_{in}$ for $v_{in} > 0$ and 0 otherwise. A2 sums v_{in} (through R) and v_A (through R/2, i.e. weight 2):

FORMULA — ABSOLUTE-VALUE CIRCUIT

$$v_o = -(v_{in} + 2v_A):\quad v_{in} > 0 \Rightarrow -(v_{in} - 2v_{in}) = v_{in};\quad v_{in} < 0 \Rightarrow -v_{in} \quad \therefore\ v_o = |v_{in}|$$

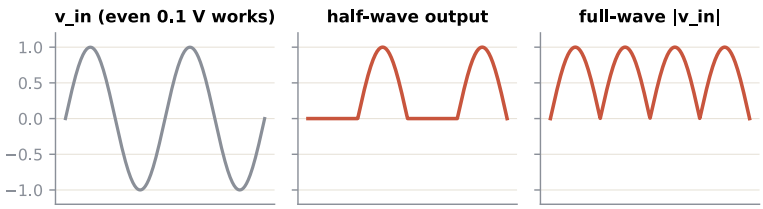


Fig 2.14 Input, precision half-wave output (Fig 2.12) and full-wave output (Fig 2.13).

2.8 Comparators

A **comparator** is an op-amp used *without* feedback. It compares v_{in} with a reference V_{ref} and, thanks to the ±65 µV window, the output is always at $+V_{sat}$ or $-V_{sat}$ — a 1-bit ADC.

- **Non-inverting comparator:** v_{in} on +, V_{ref} on −: $v_o = +V_{sat}$ when $v_{in} > V_{ref}$. **Inverting:** swap the inputs, output flips.
- **Zero-crossing detector:** $V_{ref} = 0$ — converts a sine into a square wave of the same frequency.
- Output levels can be clamped with back-to-back Zeners; dedicated comparators (LM311, LM339) have open-collector outputs and much faster response than a 741 (whose 0.5 V/µs slew makes a 26 V swing take 52 µs).

The problem: a slow, noisy input lingering near V_{ref} crosses it many times, and the output **chatters** (Fig 2.17, right). Cure: positive feedback — the Schmitt trigger.

2.9 Schmitt trigger

A Schmitt trigger is a comparator with **positive feedback**: part of the output is fed back to the + input. The threshold therefore depends on the present output state — there are two thresholds, an upper V_{UT} and a lower V_{LT}. Once the output switches, the input must travel all the way to the *other* threshold to switch it back. Noise smaller than the gap (the **hysteresis** V_H) is simply ignored.

> **THINK OF IT LIKE**
>
> A room thermostat that turns the AC on at 26 °C but off only at 23 °C. Without that 3 °C gap it would click on and off every few seconds.

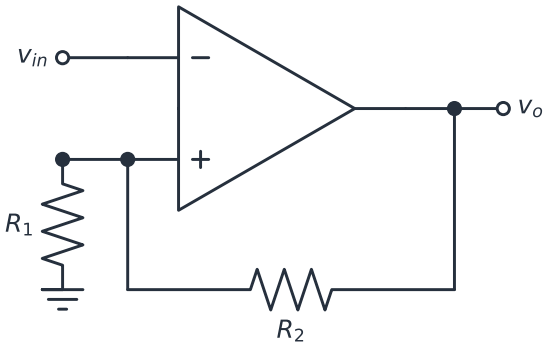


Fig 2.15 Inverting Schmitt trigger.

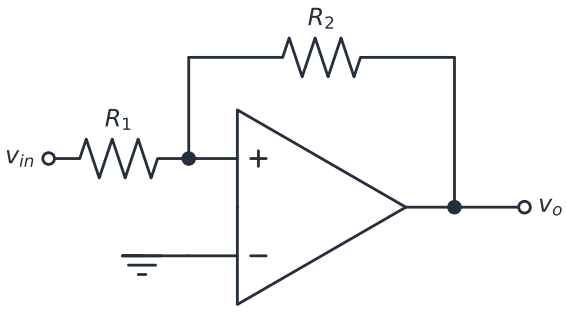


Fig 2.16 Non-inverting Schmitt trigger.

Inverting (Fig 2.15). The + input is $v_o R_1/(R_1 + R_2)$. With $v_o = +V_{sat}$ this is V_{UT}; when v_{in} rises past it, v_o snaps to $-V_{sat}$ and the threshold drops to V_{LT}. If R_1 returns to V_{ref} instead of ground, both thresholds shift.

Non-inverting (Fig 2.16). Switching occurs when $v_+ = 0$: $v_{in}R_2/(R_1 + R_2) + v_o R_1/(R_1 + R_2) = 0 \Rightarrow v_{in} = -v_o R_1/R_2$.

FORMULA — SCHMITT THRESHOLDS ($\pm V_{SAT}$ OUTPUTS)

$$\text{Inverting: } V_{UT} = \frac{R_1}{R_1 + R_2} V_{sat}, \quad V_{LT} = -\frac{R_1}{R_1 + R_2} V_{sat}, \quad V_H = \frac{2R_1 V_{sat}}{R_1 + R_2}$$

$$\text{With } V_{ref} \text{ on } R_1\text{: } V_{UT,LT} = \frac{R_2 V_{ref} \pm R_1 V_{sat}}{R_1 + R_2} \qquad \text{Non-inverting: } V_{UT,LT} = \pm \frac{R_1}{R_2} V_{sat}$$

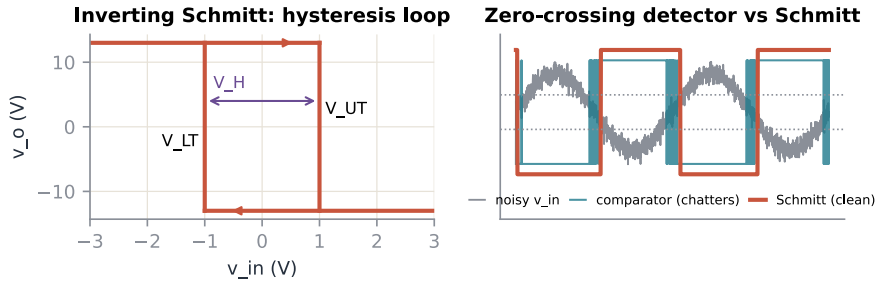


Fig 2.17 Left: hysteresis loop (V_{UT} = +1 V, V_{LT} = −1 V, Example 2.8). Right: a noisy input makes a plain comparator chatter; the Schmitt output is clean.

SOLVED EXAMPLE 2.8 · SCHMITT DESIGN (KTU FAVOURITE)

An inverting Schmitt trigger uses R_1 = 10 kΩ, R_2 = 120 kΩ, V_{sat} = ±13 V. (a) Find V_{UT}, V_{LT}, V_H. (b) Repeat if R_1 is returned to V_{ref} = +2 V.

- (a) $V_{UT} = 13 \times 10/130 = +1$ V, $V_{LT} = -1$ V, $V_H = 2$ V
- (b) $V_{UT} = (120\text{ k} \times 2 + 10\text{ k} \times 13)/130\text{ k} = 370/130 = 2.85$ V
- $V_{LT} = (240 - 130)/130 = 0.85$ V; V_H still 2 V (the loop just shifts right by 1.85 V)

(a) ±1 V, V_H = 2 V (b) 2.85 V and 0.85 V, V_H = 2 V

EXAM TIP

Always draw the hysteresis loop with arrows: for an inverting Schmitt the output is HIGH on the left, the loop is traversed anticlockwise when you sweep up then down. For a non-inverting Schmitt it's the mirror image (LOW on the left).

2.10 Log and antilog amplifiers

A diode's current is exponential in its voltage: $I_D = I_S e^{V_D/V_T}$. Turn it around and $V_D = V_T \ln(I_D/I_S)$ — a logarithm for free. Place the diode (or a grounded-base transistor, which is more ideal) where the op-amp forces a known current through it.

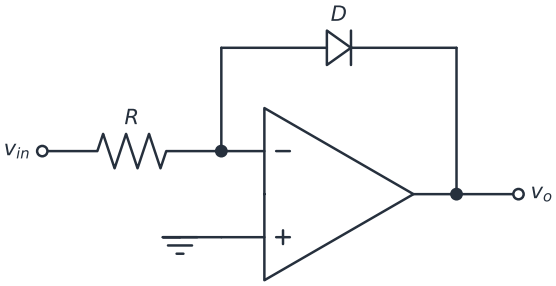


Fig 2.18 Log amplifier.

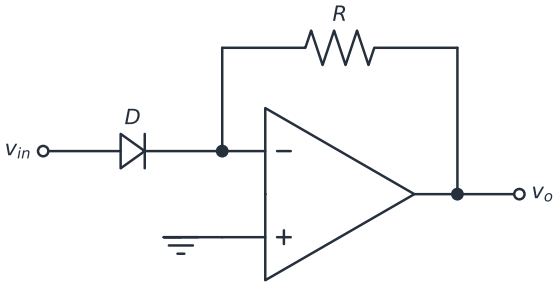


Fig 2.19 Antilog amplifier.

FORMULA

Log: $i = v_{in}/R$ through D $\Rightarrow v_o = -V_T \ln \frac{v_{in}}{RI_S}$ ($v_{in} > 0$ only) Antilog: $v_o = -RI_S\, e^{v_{in}/V_T}$

Both V_T and I_S depend strongly on temperature. Practical log amps use a **matched transistor pair** (the I_S terms cancel in a ratio) and a temperature-compensating resistor (thermistor) to cancel V_T. Uses: compressing a huge dynamic range (dB meters, audio), and analog multiply/divide (log → add → antilog), RMS converters.

SOLVED EXAMPLE 2.9 · GATE-STYLE

A log amplifier has R = 10 kΩ, $I_S = 10^{-13}$ A, V_T = 26 mV. Find v_o for v_{in} = 1 V. How much does v_o change when v_{in} becomes 10 V?

- $v_{in}/(RI_S) = 1/(10^4 \times 10^{-13}) = 10^9 \Rightarrow v_o = -0.026 \times \ln 10^9 = -0.026 \times 20.72 = -0.539$ V
- A ×10 change adds $V_T \ln 10 = 0.026 \times 2.303 = 59.9$ mV

$v_o \approx -0.539$ V; ≈ −60 mV more per decade (→ −0.599 V)

2.11 Peak detector and sample-and-hold

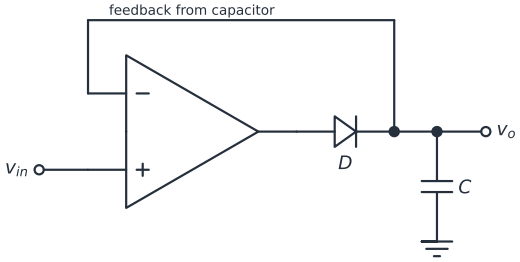


Fig 2.20 Positive peak detector.

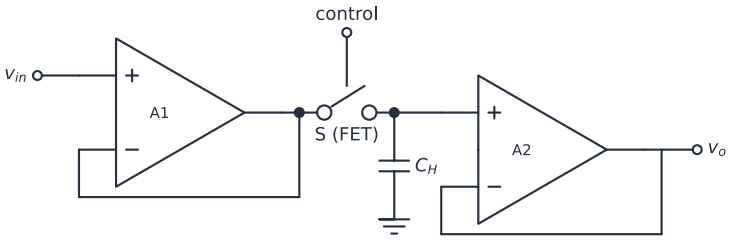


Fig 2.21 Sample-and-hold with two followers.

Peak detector (Fig 2.20). While v_{in} is above the capacitor voltage, the op-amp output rises, D conducts and charges C until $v_C = v_{in}$ — feedback is taken from the capacitor, so the diode drop is cancelled. When v_{in} falls, D reverse-biases and C **holds the highest value** seen. Reverse the diode for a negative peak detector. In practice a buffer follows C (to stop the load discharging it) and a switch across C resets it. Used in AM detection, test equipment, sound-level meters.

Sample-and-hold (Fig 2.21). When the control is high, the FET switch is ON and C_H tracks v_{in} through the low-resistance buffer A1 (**sample**). When the control goes low the switch opens and C_H keeps the last value; A2 (a high-input-impedance follower) reads it without discharging it (**hold**). Every ADC needs one so the input doesn't change mid-conversion. IC LF398 is the classic part.

Specs: *acquisition time* (time for C_H to charge to within the error band after sample starts), *aperture time* (delay between the hold command and the switch actually opening), *droop rate* = I_{leak}/C_H during hold (so use low-leakage polystyrene/Teflon capacitors), and *feedthrough* (input leaking to the output while holding). Choosing C_H is a trade-off: large C means slow droop but long acquisition time. Example: a 1 nA leakage on 0.01 µF droops at $10^{-9}/10^{-8} = 0.1$ V/s.

INTERVIEW CORNER

Q. Why is positive feedback used in a Schmitt trigger but negative feedback in an amplifier?
A. Positive feedback makes switching regenerative and fast and creates two stable states with hysteresis; negative feedback keeps the op-amp in its linear region.

Q. Why is an ideal differentiator rarely used?
A. Its gain rises with frequency, so it amplifies noise and is prone to instability; its input impedance ($1/j\omega C$) also falls to zero at high frequency.

Q. What does the op-amp do in a precision rectifier?
A. It divides the diode's cut-in voltage by the open-loop gain, so millivolt signals get rectified accurately.

✓ Module 2 — Quick revision

Rules: $v_+ = v_-$; no input current. **Inverting:** $-R_f/R_1$, $R_{in} = R_1$. **Non-inv:** $1 + R_f/R_1$. **Follower:** 1

Feedback: $A_F = A/(1 + A\beta)$; $R_{iF} = R_i(1 + A\beta)$; $R_{oF} = R_o/(1 + A\beta)$; $f_F = f_o(1 + A\beta)$

Summer: $v_o = -R_f\Sigma\, v_k/R_k$. **Difference:** $v_o = (R_2/R_1)(v_2 - v_1)$

IA: $v_o = (R_4/R_3)(1 + 2R/R_G)(v_2 - v_1)$

Integrator: $v_o = -(1/R_1C_f)\int v_{in}dt$; $f_a = 1/2\pi R_fC_f$, $f_b = 1/2\pi R_1C_f$

Differentiator: $v_o = -R_fC_1dv_{in}/dt$; $f_a = 1/2\pi R_fC_1$, $f_b = 1/2\pi R_1C_1 = 10–20\ f_a$

V→I: $i_L = v_{in}/R_1$. **I→V:** $v_o = -i_{in}R_f$

Precision rectifier: cut-in = 0.7/A. FWR: $v_o = -(v_{in} + 2v_A) = |v_{in}|$

Schmitt (inv): $V_{UT,LT} = \pm R_1V_{sat}/(R_1+R_2)$; $V_H = 2R_1V_{sat}/(R_1+R_2)$

Schmitt (non-inv): $\pm(R_1/R_2)V_{sat}$

Log: $v_o = -V_T\ln(v_{in}/RI_S)$ (60 mV/decade). **Antilog:** $-RI_Se^{v_{in}/V_T}$

S/H: droop = I_{leak}/C_H; acquisition vs aperture time

Practice — short answer (3 marks)

1. Derive the gain of a non-inverting amplifier. Why is a follower called a buffer?
2. What is a virtual ground?
3. Design an averaging amplifier for three inputs.
4. Why is a practical integrator preferred over the ideal one?
5. Draw a V-to-I converter with floating load and derive i_L.
6. Compare a comparator and a Schmitt trigger.
7. Explain how a precision rectifier rectifies millivolt signals.
8. Define acquisition time, aperture time and droop rate.

Practice — long answer (9 / 14 marks)

1. Explain the effect of negative feedback on gain, R_i, R_o and bandwidth of a non-inverting amplifier.
2. Derive the output of a three-op-amp instrumentation amplifier. Design it for gain 50 with R = 10 kΩ, $R_3 = R_4$. [$R_G \approx 408\ \Omega$]
3. Explain the practical differentiator; design for f_a = 500 Hz with C_1 = 0.1 µF, $f_b = 10f_a$. [$R_f \approx 3.18$ kΩ, $R_1 \approx 318\ \Omega$, C_f = 10 nF]
4. Explain precision half-wave and full-wave rectifiers with waveforms.
5. Derive V_{UT}, V_{LT} of an inverting Schmitt trigger. Design for V_{UT} = 2 V, V_{LT} = −2 V with V_{sat} = ±12 V. [$R_2 = 5R_1$]
6. Explain log/antilog amplifiers and the sample-and-hold circuit.

03

Oscillators, Waveform Generators, Active Filters & IC 723

> *Every clock, every audio equaliser and every bench power supply lives here. Filter and oscillator design questions are KTU regulars, and "why does an oscillator start?" is a classic interview opener.*

SYLLABUS CHECKLIST

- ☐ Barkhausen criterion; RC phase-shift and Wien bridge oscillators (derivations)
- ☐ Square, triangular and sawtooth wave generators
- ☐ Astable and monostable multivibrators using op-amp
- ☐ First and second order Butterworth low-pass and high-pass filters (design)
- ☐ Band-pass, band-reject and all-pass filters
- ☐ IC 723 — internal diagram, low- and high-voltage regulators, current limiting and foldback

≈ 9 hours · CO3 · Design oscillators, waveform generators, active filters and regulators

3.1 Oscillators and the Barkhausen criterion

An **oscillator** produces a periodic output with *no input* — just DC power. It is an amplifier whose output is fed back to its own input **in phase**. Tiny noise at the right frequency goes round the loop, comes back slightly bigger, goes round again… and grows into a steady sine wave.

> **THINK OF IT LIKE**
>
> Pushing a child on a swing. Push at the right moment (in phase) with just enough force to cover the friction losses (loop gain = 1) and the swing keeps going forever.

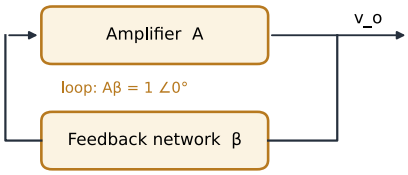


Fig 3.1 Positive-feedback loop.

BARKHAUSEN CRITERION

$$|A\beta| = 1 \quad \text{and} \quad \angle A\beta = 0^\circ \ (\text{or } 360^\circ)$$

Total phase shift round the loop must be 0°/360°, and the loop gain magnitude must be exactly 1 for steady oscillation. In practice we start with $|A\beta|$ slightly **greater** than 1 so oscillations build up; the amplifier's non-linearity (or an amplitude-control element) then pulls the effective gain back to 1.

3.2 RC phase-shift oscillator

An inverting amplifier gives 180°. Three identical RC sections, each giving about 60°, add the other 180° at exactly one frequency — total 360°. (Why three? One RC section can give at most 90°, which it reaches only when its output is zero, so at least three are needed.)

Derivation (outline). Write mesh equations for the three-section ladder (series C, shunt R) driven by v_o. With $\alpha = 1/(\omega RC)$ the feedback factor works out to

$$\beta = \frac{v_f}{v_o} = \frac{1}{(1 - 5\alpha^2) - j(6\alpha - \alpha^3)}$$

Phase of $\beta = -180^\circ$ (real and negative) needs the imaginary part to vanish: $6\alpha - \alpha^3 = 0 \Rightarrow \alpha^2 = 6 \Rightarrow \omega = 1/(\sqrt{6}\,RC)$

At this frequency $\beta = 1/(1 - 30) = -1/29$, so $|A|$ must be at least 29.

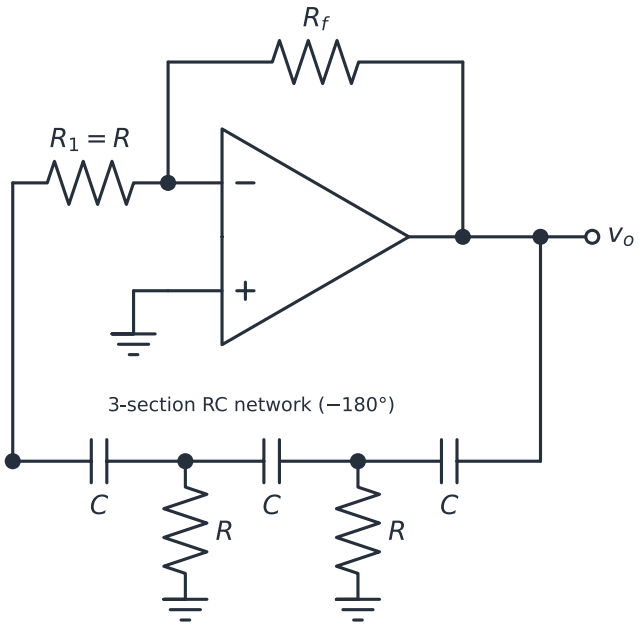


Fig 3.2 RC phase-shift oscillator. The third R is R_1, returned to the virtual ground.

FORMULA — RC PHASE-SHIFT

$$f_o = \frac{1}{2\pi\sqrt{6}\,RC} \qquad |\beta| = \frac{1}{29} \Rightarrow R_f \geq 29R_1 \;\; (R_1 = R)$$

SOLVED EXAMPLE 3.1

Design an RC phase-shift oscillator for f_o = 1 kHz using C = 0.01 µF.

- $R = 1/(2\pi\sqrt{6}\, f_o C) = 1/(2\pi \times 2.449 \times 1000 \times 10^{-8}) = 6.50$ kΩ
- $R_1 = R = 6.5$ kΩ (to avoid loading the network, $R_1 \geq 10R$ is preferred in theory; KTU accepts $R_1 = R$)
- $R_f \geq 29 \times 6.5$ k = 188.5 kΩ → use a 200 kΩ (or 250 kΩ pot) to ensure start-up

$R = R_1 \approx 6.5$ kΩ, $R_f \approx 200$ kΩ (f_o = 1.0 kHz)

3.3 Wien bridge oscillator

Here a **non-inverting** amplifier (0°) is used, so the feedback network must also give 0°. The Wien network — a series RC (Z_1) followed by a parallel RC (Z_2) — does exactly that at one frequency. It's the standard low-distortion audio oscillator (10 Hz – 1 MHz).

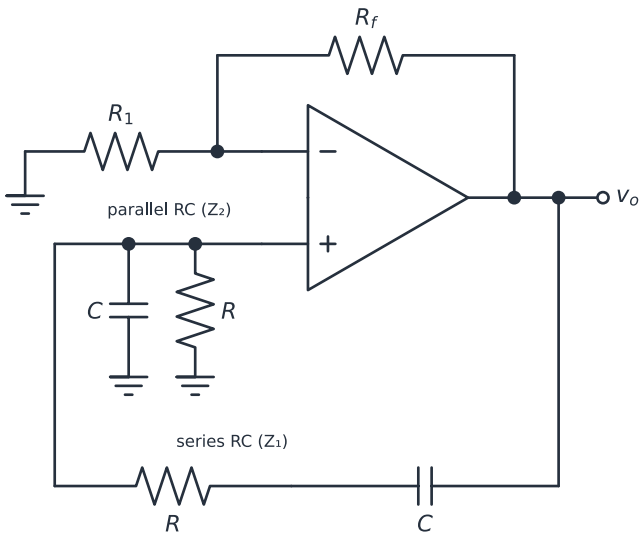


Fig 3.3 Wien bridge oscillator: R_f–R_1 set the gain, the Wien network sets the frequency.

$Z_1 = R + 1/(j\omega C)$, $Z_2 = R/(1 + j\omega RC)$. $\beta = Z_2/(Z_1 + Z_2)$; multiplying out:

$$\beta = \frac{1}{3 + j\left(\omega RC - \frac{1}{\omega RC}\right)}$$

β is real (0°) when $\omega RC = 1/(\omega RC) \Rightarrow \omega_o = 1/RC$. There $\beta = 1/3$, so the amplifier needs $A = 1 + R_f/R_1 = 3$.

FORMULA — WIEN BRIDGE

$$f_o = \frac{1}{2\pi RC} \qquad \beta = \frac{1}{3} \Rightarrow R_f = 2R_1 \text{ (slightly more to start)}$$

Amplitude stabilisation: with gain exactly 3 the oscillation would die or grow. Practical circuits make R_1 a small lamp (or R_f a thermistor, or add back-to-back diodes across part of R_f) so the gain falls automatically as the amplitude rises.

SOLVED EXAMPLE 3.2

Design a Wien bridge oscillator for 1 kHz with C = 0.01 µF and R_1 = 10 kΩ.

- $R = 1/(2\pi \times 1000 \times 10^{-8}) = 15.9$ kΩ
- $R_f = 2R_1 = 20$ kΩ (use a 25 kΩ pot set just above 20 kΩ)

R ≈ 15.9 kΩ, R_f ≈ 20 kΩ

INTERVIEW CORNER

Q. How does an oscillator start without an input?

A. Thermal noise and the power-on transient contain all frequencies; the loop selects the one meeting the phase condition and amplifies it ($|A\beta| > 1$) until non-linearity limits it.

Q. Why are RC oscillators used at low frequencies and LC at high?

A. At MHz, R and C values become tiny and op-amp phase shift spoils the condition; at audio, L would be huge and bulky, so RC is preferred.

3.4 Square-wave generator (astable multivibrator)

An **astable** has no stable state — it keeps flipping between $+V_{sat}$ and $-V_{sat}$. Build it from an inverting Schmitt trigger whose input is a capacitor charged from its own output.

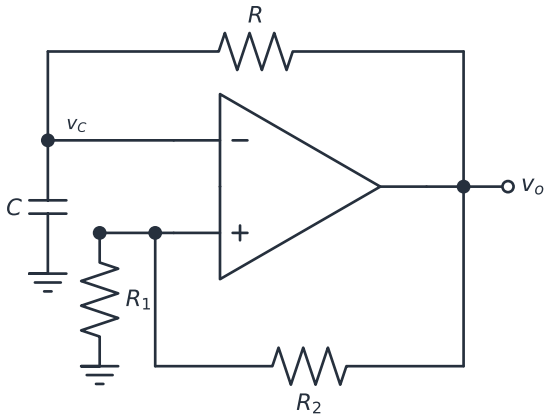


Fig 3.4 Op-amp astable.

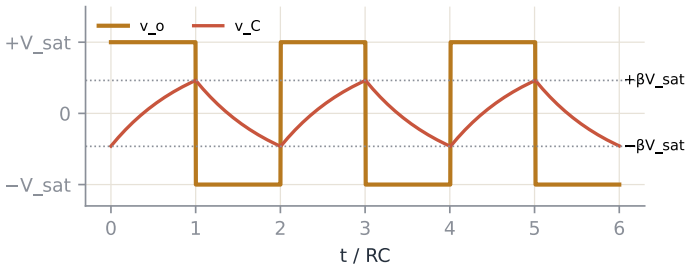


Fig 3.5 Output and capacitor voltage.

Working. Let $v_o = +V_{sat}$. Then $v_+ = \beta V_{sat}$ with $\beta = R_1/(R_1 + R_2)$, and C charges through R towards $+V_{sat}$. The instant v_C crosses βV_{sat}, the output flips to $-V_{sat}$, v_+ becomes $-\beta V_{sat}$, and C discharges towards $-V_{sat}$... and so on.

Charging from $-\beta V_{sat}$ towards $+V_{sat}$: $v_C(t) = V_{sat} - V_{sat}(1 + \beta)e^{-t/RC}$

Set $v_C(T_1) = \beta V_{sat} \Rightarrow e^{-T_1/RC} = (1 - \beta)/(1 + \beta) \Rightarrow T_1 = RC \ln[(1 + \beta)/(1 - \beta)]$; the other half is identical.

FORMULA — ASTABLE

$$T = 2RC \ln \frac{1+\beta}{1-\beta}, \quad \beta = \frac{R_1}{R_1 + R_2} \qquad \text{If } R_2 = 1.16R_1\text{: } T = 2RC,\ f = 1/(2RC)$$

SOLVED EXAMPLE 3.3

Design a 1 kHz square-wave generator with C = 0.1 µF and R_1 = 10 kΩ.

- Choose $R_2 = 1.16R_1 = 11.6$ kΩ $\Rightarrow \beta = 1/2.16 = 0.463$, $\ln(1.463/0.537) = 1.002 \approx 1 \Rightarrow T = 2RC$
- $R = T/(2C) = 1\text{ ms}/(2 \times 0.1\ \mu\text{F}) = 5$ kΩ

R = 5 kΩ, R_1 = 10 kΩ, R_2 = 11.6 kΩ

3.5 Triangular and sawtooth wave generators

Integrate a square wave and you get a triangle (Module 2). The neat trick is to let the triangle drive the comparator that makes the square wave — a closed loop of a **non-inverting Schmitt (A1)** and an **integrator (A2)**. This gives a much more linear triangle than the RC exponential of the astable.

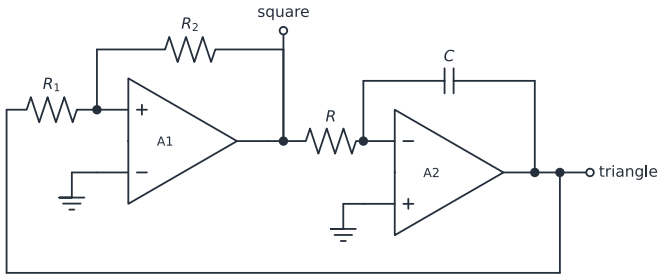


Fig 3.6 Triangular-wave generator: A1 gives the square, A2 the triangle.

The integrator ramps at V_{sat}/RC. A1 switches when the triangle reaches $\pm(R_1/R_2)V_{sat}$ (its thresholds). Each half-period the triangle travels $2(R_1/R_2)V_{sat}$:

$(V_{sat}/RC) \times T/2 = 2(R_1/R_2)V_{sat} \Rightarrow T = 4RC{\cdot}R_1/R_2$

FORMULA — TRIANGULAR GENERATOR

$$f = \frac{R_2}{4R_1RC} \qquad V_{o(pp)} = 2\frac{R_1}{R_2}V_{sat}$$

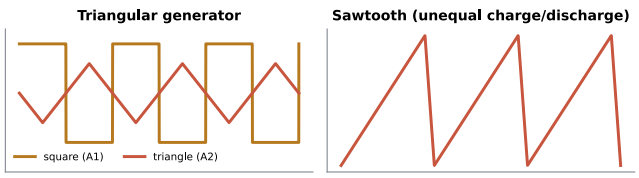


Fig 3.7 Square + triangle from Fig 3.6, and a sawtooth.

Sawtooth. A sawtooth is just a triangle whose rise and fall times are very unequal. Make the integrator charge slowly and discharge quickly — e.g. add a diode with a small resistor in parallel with R so one direction sees a much smaller resistance, or add a DC offset (pot) at the integrator's + input so the two ramps have different slopes.

SOLVED EXAMPLE 3.4

In Fig 3.6, R_1 = 10 kΩ, R_2 = 20 kΩ, R = 10 kΩ, C = 0.05 µF, V_{sat} = ±13 V. Find f and the triangle's peak-to-peak value.

▸ f = 20 k/(4 × 10 k × 10 k × 0.05 µ) = 20 000/20 = 1000 Hz

▸ $V_{pp} = 2 \times (10/20) \times 13 = 13$ V (i.e. ±6.5 V)

f = 1 kHz, triangle = 13 V peak-to-peak

3.6 Monostable multivibrator

A **monostable** ("one-shot") has one stable state. A trigger pulse kicks it into the other (quasi-stable) state for a fixed time T, then it returns on its own. Uses: pulse stretching, debouncing, timing delays.

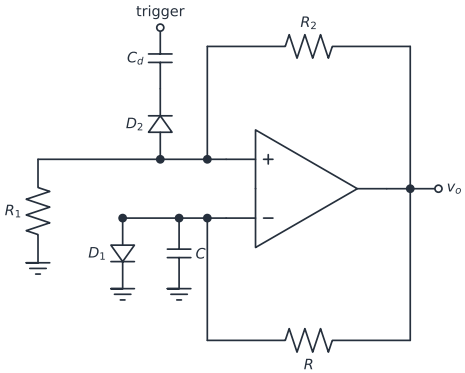


Fig 3.8 Op-amp monostable.

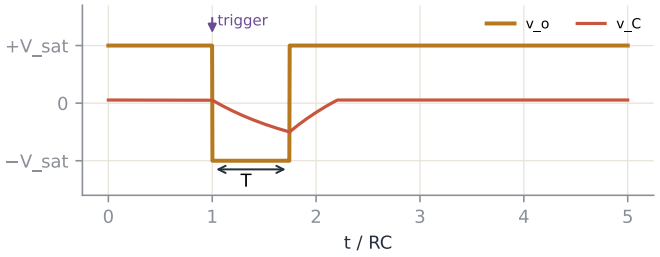


Fig 3.9 Waveforms (β = 0.5).

Stable state: $v_o = +V_{sat}$; D_1 clamps the capacitor at $+V_D$ (≈ 0.7 V), while $v_+ = \beta V_{sat}$ is larger, so the output stays high. **Trigger:** a negative pulse through C_d and D_2 pulls v_+ below V_D; the output flips to $-V_{sat}$, $v_+ = -\beta V_{sat}$. C now charges from $+V_D$ towards $-V_{sat}$; when it reaches $-\beta V_{sat}$ the output flips back. C then recovers to V_D (recovery time) before the next trigger.

$v_C(t) = -V_{sat} + (V_{sat} + V_D)e^{-t/RC} = -\beta V_{sat}$ at $t = T$

FORMULA — MONOSTABLE PULSE WIDTH

$$T = RC \ln \frac{1 + V_D/V_{sat}}{1 - \beta} \approx RC \ln \frac{1}{1 - \beta} \quad (\beta = 0.5 \Rightarrow T \approx 0.69RC)$$

SOLVED EXAMPLE 3.5

A monostable has R = 10 kΩ, C = 0.1 µF, $R_1 = R_2$, V_{sat} = 13 V, V_D = 0.7 V. Find T.

▸ β = 0.5; T = 1 ms × ln[(1 + 0.7/13)/0.5] = 1 ms × ln(2.108) = 0.746 ms

T ≈ 0.75 ms (0.69 ms if V_D is neglected)

3.7 Active filters and first-order Butterworth LPF/HPF

A **filter** passes some frequencies and blocks others. **Active** filters use op-amps with R and C (no bulky inductors), can give gain, and have high input / low output impedance so stages cascade without loading. Their limitation is the op-amp's bandwidth (useful to a few hundred kHz).

The **Butterworth** response is *maximally flat* in the pass band — no ripple — and rolls off at −20n dB/decade for an n-th order filter. At the cut-off frequency the gain is 0.707 A_F (−3 dB).

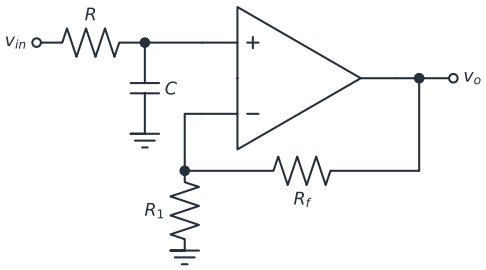


Fig 3.10 First-order LPF.

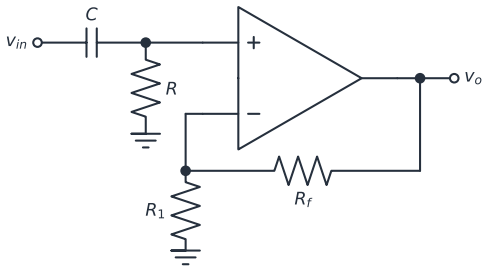


Fig 3.11 First-order HPF (R and C swapped).

The RC divider at the + input sets the shape, the non-inverting amplifier sets the pass-band gain $A_F = 1 + R_f/R_1$.

FORMULA — FIRST ORDER

$$\text{LPF: } \frac{v_o}{v_{in}} = \frac{A_F}{1 + j(f/f_H)} \qquad \text{HPF: } \frac{v_o}{v_{in}} = \frac{A_F\, j(f/f_L)}{1 + j(f/f_L)} \qquad f_H,\ f_L = \frac{1}{2\pi RC}$$

EXAM TIP — THE 4-STEP DESIGN RECIPE (ANY ORDER)

(1) Choose the cut-off frequency. (2) Pick C ≤ 1 µF (e.g. 0.01 µF). (3) Compute $R = 1/(2\pi f_c C)$. (4) Choose R_1 (10 kΩ) and get R_f from the required A_F. To move the cut-off later (**frequency scaling**): new R = old R × (old f_c/new f_c), keep C.

3.8 Second-order Butterworth LPF/HPF

A second RC section plus a feedback capacitor (the **Sallen–Key** or VCVS topology) gives −40 dB/decade. For a Butterworth response the damping must be $\sqrt{2} = 1.414$; with equal R's and C's this fixes the gain: $A_F = 3 - 1.414 =$ **1.586**, i.e. $R_f = 0.586R_1$.

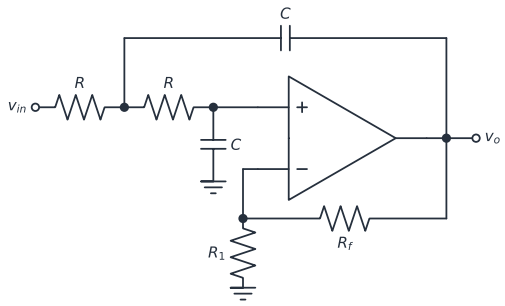


Fig 3.12 Second-order LPF (Sallen–Key).

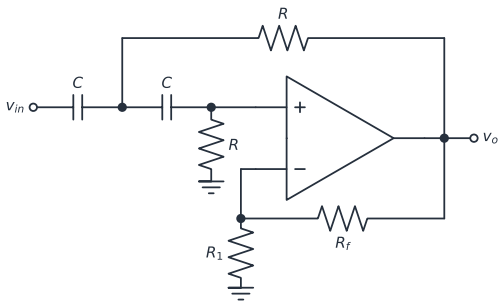


Fig 3.13 Second-order HPF.

FORMULA — SECOND ORDER (EQUAL R, C)

$$|v_o/v_{in}| = \frac{A_F}{\sqrt{(1+(f/f_H)^4)}}\ \text{(LPF)},\quad \frac{A_F(f/f_L)^2}{\sqrt{(1+(f/f_L)^4)}}\ \text{(HPF)},\quad f_c = \frac{1}{2\pi RC},\quad A_F = 1.586$$



Fig 3.14 Butterworth responses (dotted line = −3 dB).

SOLVED EXAMPLE 3.6 · SECOND-ORDER LPF DESIGN

Design a second-order Butterworth LPF with f_H = 1 kHz using C = 0.0047 µF. Then scale it to 2 kHz.

- R = 1/(2π × 1000 × 0.0047 µ) = 33.9 kΩ (use 33 kΩ + trim)
- A_F = 1.586: take R_1 = 10 kΩ ⇒ R_f = 5.86 kΩ
- Scaling to 2 kHz: R′ = 33.9 k × (1/2) = 16.9 kΩ, C unchanged

R = 33.9 kΩ, R_1 = 10 kΩ, R_f = 5.86 kΩ; R′ = 16.9 kΩ for 2 kHz

3.9 Band-pass and band-reject filters

A band-pass filter is described by its centre frequency $f_c = \sqrt{(f_L f_H)}$, bandwidth BW = $f_H - f_L$ and quality factor Q = f_c/BW. **Wide-band** (Q < 10): cascade an HPF (sets f_L) and an LPF (sets f_H); total gain = product of gains. **Narrow-band** (Q > 10, or a sharp peak): use the multiple-feedback (MFB) circuit of Fig 3.15.

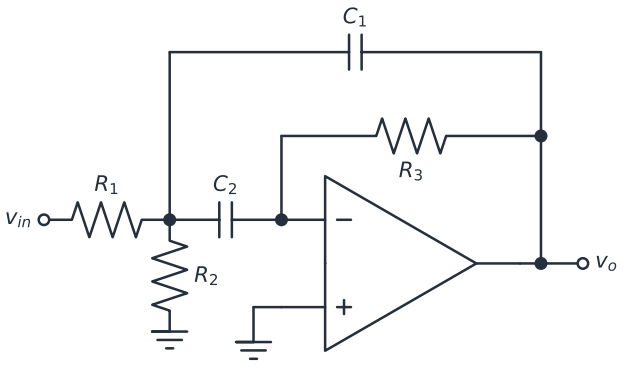


Fig 3.15 Narrow-band MFB band-pass ($C_1 = C_2 = C$).

FORMULA — MFB BAND-PASS

$$R_1 = \frac{Q}{2\pi f_c C A_F}, \quad R_2 = \frac{Q}{2\pi f_c C(2Q^2 - A_F)}$$

$$R_3 = \frac{Q}{\pi f_c C}, \quad A_F = \frac{R_3}{2R_1} < 2Q^2$$

Bonus: f_c can be moved without changing gain or BW by changing R_2 alone: $R_2' = R_2(f_c/f_c')^2$.

Band-reject (band-stop). *Wide-band:* feed the input to an HPF and an LPF in parallel and add their outputs with a summing amplifier; the LPF cut-off must be **below** the HPF cut-off ($f_H < f_L$) to leave a stop band. *Narrow-band (notch):* the passive **twin-T** network (two R's and a 2C in one T, two C's and an R/2 in the other) has a deep null at $f_N = 1/(2\pi RC)$; following it with a voltage follower (and bootstrapping the T's common point) raises its Q. Classic use: removing 50 Hz mains hum from ECG and audio.

SOLVED EXAMPLE 3.7 · NARROW BAND-PASS DESIGN

Design an MFB band-pass filter with f_c = 1 kHz, Q = 3, A_F = 10, C = 0.01 µF. Then move f_c to 1.5 kHz.

- Check $A_F < 2Q^2 = 18$ √. BW = f_c/Q = 333 Hz
- $R_1 = 3/(2\pi \times 1000 \times 10^{-8} \times 10) = 4.77$ kΩ
- $R_2 = 3/(2\pi \times 1000 \times 10^{-8} \times (18 - 10)) = 5.97$ kΩ; $R_3 = 3/(\pi \times 1000 \times 10^{-8}) = 95.5$ kΩ
- At 1.5 kHz: $R_2' = 5.97\text{ k} \times (1/1.5)^2 = 2.65$ kΩ

R_1 = 4.77 kΩ, R_2 = 5.97 kΩ, R_3 = 95.5 kΩ; R_2' = 2.65 kΩ

SOLVED EXAMPLE 3.8 · WIDE-BAND FILTERS

(a) A wide-band BPF has f_L = 200 Hz and f_H = 1 kHz. Find f_c and Q. (b) Find R for a 50 Hz twin-T notch with C = 0.068 µF.

- (a) $f_c = \sqrt{(200 \times 1000)} = 447$ Hz; Q = 447/800 = 0.56 (< 10, so cascading is right)
- (b) $R = 1/(2\pi \times 50 \times 0.068\ \mu) = 46.8$ kΩ (so R/2 = 23.4 kΩ, 2C = 0.136 µF)

(a) f_c ≈ 447 Hz, Q ≈ 0.56 (b) R ≈ 46.8 kΩ

3.10 All-pass filter

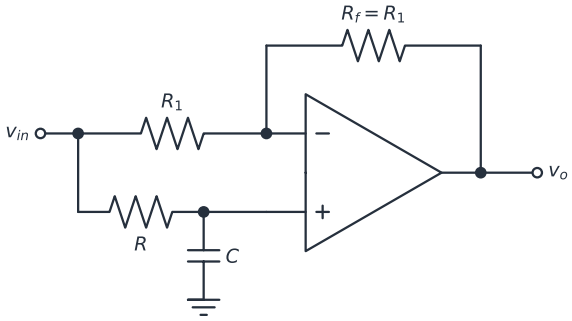


Fig 3.16 First-order all-pass (phase shifter).

An **all-pass** filter passes every frequency with the **same gain** but shifts the **phase** by an amount that depends on frequency. It is used for phase correction (delay equalisation) in communication links and as an adjustable phase shifter.

By superposition, $v_o = -v_{in} + 2v_+$ with $v_+ = v_{in}/(1 + j\omega RC)$:

FORMULA

$$\frac{v_o}{v_{in}} = \frac{1 - j\omega RC}{1 + j\omega RC}, \quad |H| = 1, \quad \varphi = -2\tan^{-1}(2\pi fRC)$$

Example: R = 15.9 kΩ, C = 0.01 µF gives $\varphi = -2\tan^{-1}(1) = -90°$ at 1 kHz; φ sweeps from 0° (DC) to −180° (very high f). Swapping R and C gives a leading phase.

3.11 IC 723 voltage regulator

A **voltage regulator** holds its output constant despite changes in input voltage or load current. The 723 is a general-purpose adjustable regulator: V_{in} 9.5–40 V, V_o 2–37 V, up to 150 mA directly (amps with an external pass transistor), with an internal 7.15 V reference.

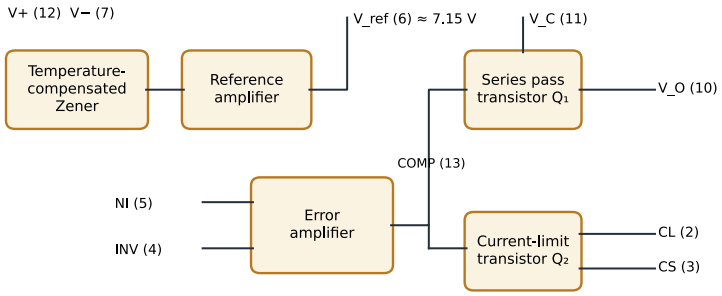


Fig 3.17 Functional diagram of the 723 (14-pin DIP numbers in brackets).

Inside: a temperature-compensated Zener feeds a reference amplifier to give a stable $V_{ref} \approx 7.15$ V (pin 6). The **error amplifier** compares a fraction of the output (INV, pin 4) with the reference (NI, pin 5) and drives the **series pass transistor** Q_1, which drops the excess voltage. Q_2 (CL, CS pins) turns on when the load current is too high and steals Q_1's base drive. A 100 pF capacitor from COMP (13) to INV (4) stabilises the loop.

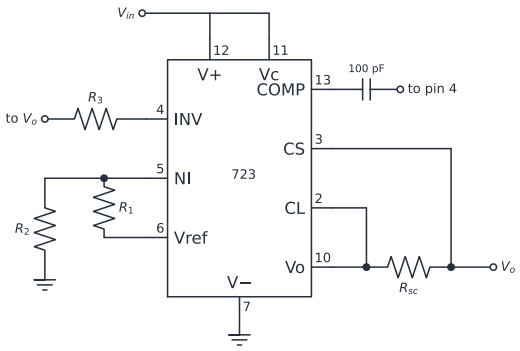


Fig 3.18 Low-voltage regulator (2–7 V).

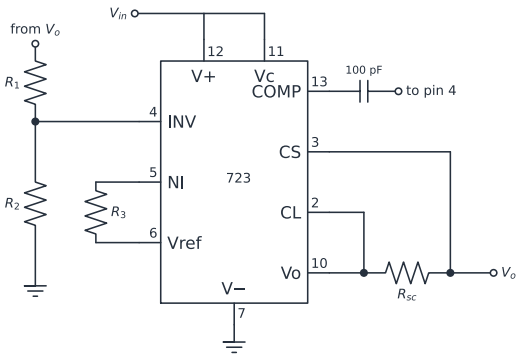


Fig 3.19 High-voltage regulator (7–37 V).

FORMULA — 723 OUTPUT VOLTAGE

Low ($V_o < 7$ V): $V_o = V_{ref}\dfrac{R_2}{R_1 + R_2}$, $R_3 = R_1 \parallel R_2$ High ($V_o > 7$ V): $V_o = V_{ref}(1 + \dfrac{R_1}{R_2})$, $R_3 = R_1 \parallel R_2$

Current limiting and foldback

Simple current limit: R_{sc} sits between pin 10 and the load, with CL and CS across it. When $I_L R_{sc}$ reaches Q_2's V_{BE} ($V_{sense} \approx 0.5$ V; datasheet ≈ 0.65 V), Q_2 conducts and holds I_L at V_{sense}/R_{sc}. But under a short the pass transistor dissipates $V_{in} \times I_{limit}$.

Foldback limiting: CL is fed from a divider (R_3 from the pass-transistor side of R_{sc} to CL, R_4 from CL to ground). As overload pulls V_o down, Q_2 turns on harder and the current **folds back** to a smaller I_{sc}.

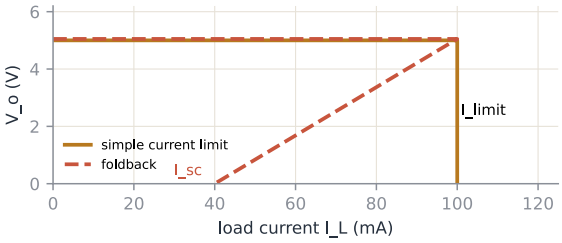


Fig 3.20 V_o–I_L: simple vs foldback (Example 3.9).

FORMULA — LIMITS

Simple: $R_{sc} = V_{sense}/I_{limit}$

Foldback: $I_{limit} = \dfrac{V_o R_3 + V_{sense}(R_3 + R_4)}{R_{sc} R_4}$

$I_{sc} = \dfrac{V_{sense}(R_3 + R_4)}{R_{sc} R_4}$

SOLVED EXAMPLE 3.9 · 723 DESIGN

5 V 723 regulator, $V_{sense} = 0.5$ V: (a) simple limit 50 mA; (b) foldback, $I_{limit} = 100$ mA, $I_{sc} = 40$ mA.

▸ Divider: $R_2/(R_1 + R_2) = 5/7.15$; with $R_1 + R_2 = 10$ kΩ ⇒ $R_2 = 6.99$ kΩ, $R_1 = 3.01$ kΩ, balancing resistor $R_1 \parallel R_2 = 2.1$ kΩ. (a) $R_{sc} = 0.5/0.05 = 10$ Ω

▸ (b) $I_{limit}/I_{sc} = 1 + (V_o/V_{sense})\cdot R_3/(R_3 + R_4) = 2.5 \Rightarrow R_3/(R_3 + R_4) = 0.15 \Rightarrow R_3 = 1.5$ kΩ, $R_4 = 8.5$ kΩ; $R_{sc} = 0.5 \times 10\text{ k}/(0.04 \times 8.5\text{ k}) = 14.7$ Ω

(a) $R_{sc} = 10$ Ω (b) $R_3 = 1.5$ kΩ, $R_4 = 8.5$ kΩ, $R_{sc} \approx 14.7$ Ω

✓ Module 3 — Quick revision

Barkhausen: $|A\beta| = 1$, $\angle A\beta = 0°/360°$; start with $|A\beta|$ slightly > 1

RC phase-shift: $f = 1/(2\pi\sqrt{6}RC)$; $\beta = 1/29 \Rightarrow R_f \geq 29R_1$

Wien bridge: $f = 1/(2\pi RC)$; $\beta = 1/3 \Rightarrow R_f = 2R_1$

Astable: $T = 2RC \ln((1+\beta)/(1-\beta))$; $R_2 = 1.16R_1 \Rightarrow T = 2RC$

Triangle: $f = R_2/(4R_1RC)$, $V_{pp} = 2(R_1/R_2)V_{sat}$. Sawtooth = unequal slopes

Monostable: $T = RC \ln((1 + V_D/V_{sat})/(1 - \beta)) \approx 0.69RC$ ($\beta = ½$)

Butterworth: maximally flat, −20n dB/dec, −3 dB at f_c

1st order: $f_c = 1/2\pi RC$, $A_F = 1 + R_f/R_1$

2nd order (equal R, C): $A_F = 1.586$, $R_f = 0.586R_1$; $|H| = A_F/\sqrt{(1 + (f/f_H)^4)}$

Scaling: $R' = R(f_c/f_c')$. **BPF:** $f_c = \sqrt{(f_L f_H)}$, $Q = f_c/BW$; wide if $Q < 10$

MFB: $R_1 = Q/(2\pi f_c C A_F)$, $R_2 = Q/(2\pi f_c C(2Q^2 - A_F))$, $R_3 = Q/(\pi f_c C)$

Notch (twin-T): $f_N = 1/2\pi RC$. **All-pass:** $\varphi = -2\tan^{-1}(2\pi fRC)$

723: $V_{ref} = 7.15$ V; low: $V_{ref}R_2/(R_1+R_2)$; high: $V_{ref}(1 + R_1/R_2)$

Limits: $R_{sc} = V_{sense}/I_{lim}$; foldback $I_{sc} = V_{sense}(R_3+R_4)/(R_{sc}R_4)$

Practice — short answer (3 marks)

1. State the Barkhausen criterion.
2. Why are three RC sections used in a phase-shift oscillator?
3. How is amplitude stabilised in a Wien bridge oscillator?
4. Design an astable for 2 kHz with C = 0.05 µF. [R = 5 kΩ, $R_2 = 1.16R_1$]
5. What is a Butterworth response? Why $A_F = 1.586$?
6. What is frequency scaling of a filter?
7. Draw the all-pass filter and derive its phase.
8. Why is foldback limiting preferred?

Practice — long answer (9 / 14 marks)

1. Derive f_o and the gain condition of an RC phase-shift oscillator.
2. Derive f_o of the Wien bridge oscillator. Design one for 500 Hz with C = 0.1 µF. [R ≈ 3.18 kΩ]
3. Explain the triangular-wave generator and derive its frequency and amplitude.
4. Explain the op-amp monostable with waveforms and derive the pulse width.
5. Design a second-order Butterworth HPF with $f_L = 5$ kHz, C = 0.01 µF. [R ≈ 3.18 kΩ, $R_f = 0.586R_1$]
6. 723 internals; design a 12 V, 100 mA-limited regulator. [$R_1/R_2 \approx 0.678$, $R_{sc} = 5$ Ω]

04

555 Timer, VCO, PLL & Data Converters

> *The 555 is the most-sold IC in history, PLLs sit inside every radio and processor clock, and DAC/ADC questions ("how does a SAR ADC work?") appear in almost every core-company interview.*

SYLLABUS CHECKLIST

- ☐ IC 555 timer — internal diagram; astable and monostable design; applications
- ☐ Voltage-controlled oscillator LM566
- ☐ PLL IC 565 — block diagram, lock and capture range
- ☐ PLL applications — FM demodulation, frequency multiplication
- ☐ DAC specifications; weighted-resistor, R-2R ladder and inverted R-2R DACs
- ☐ ADC specifications; flash, counter-type, successive-approximation and dual-slope ADCs

≈ 9 hours · CO4 · Design timer circuits and explain PLLs and data converters

4.1 IC 555 timer — inside the chip

The 555 is a tiny timing machine: two comparators watch a capacitor voltage, and a flip-flop remembers which threshold was crossed last. With one resistor–capacitor pair you get precise delays from microseconds to hours, on any supply from 4.5 V to 16 V, with 200 mA output drive.

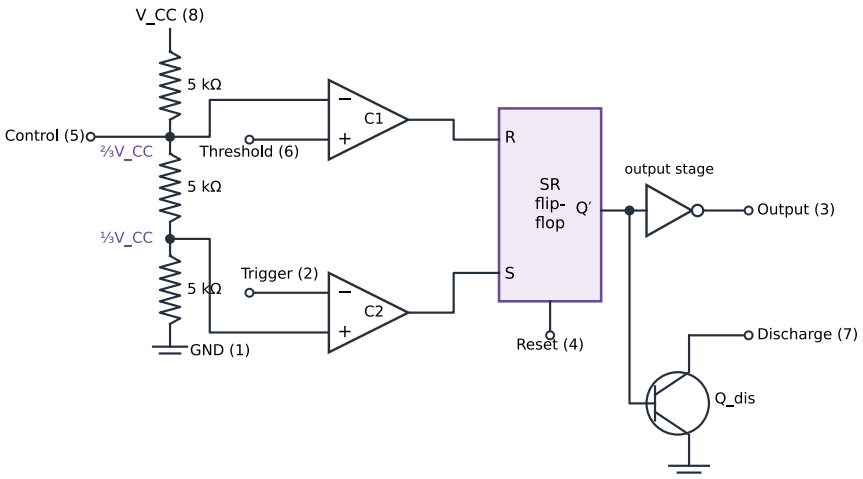


Fig 4.1 Internal diagram of the 555 (pin numbers in brackets).

- **Divider:** three equal 5 kΩ resistors set reference points at $\frac{2}{3}V_{CC}$ and $\frac{1}{3}V_{CC}$ (hence the name "555"). Pin 5 (control) gives direct access to the ⅔ point — bypass it with 0.01 µF, or drive it to change the thresholds.
- **Upper comparator C1:** when threshold (pin 6) > $\frac{2}{3}V_{CC}$, it **resets** the flip-flop → output LOW, discharge transistor ON.
- **Lower comparator C2:** when trigger (pin 2) < $\frac{1}{3}V_{CC}$, it **sets** the flip-flop → output HIGH, discharge transistor OFF.
- **Discharge (pin 7):** a transistor that shorts the timing capacitor to ground while the output is LOW. **Reset (pin 4):** active-low, overrides everything; tie to V_{CC} when unused.

4.2 555 monostable multivibrator

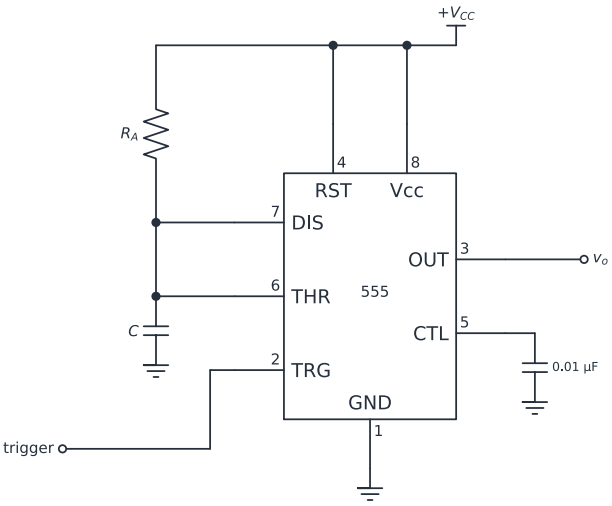


Fig 4.2 555 monostable.

Stable state: output LOW, discharge ON, C held at 0 V. **Trigger:** a negative pulse at pin 2 dipping below $\frac{1}{3}V_{CC}$ sets the flip-flop — output goes HIGH and the discharge transistor releases C, which charges through R_A towards V_{CC}. When v_C reaches $\frac{2}{3}V_{CC}$, C1 resets the flip-flop: output LOW, C is dumped. One trigger → one pulse of fixed width.

$v_C = V_{CC}(1 - e^{-t/R_AC}) = \frac{2}{3}V_{CC} \Rightarrow e^{-T/R_AC} = \frac{1}{3} \Rightarrow T = R_AC \ln 3$

FORMULA

$$T = 1.1\, R_A C$$

The trigger pulse must be **shorter** than T (otherwise use an RC differentiator at pin 2). T is independent of V_{CC}, because both the charging target and the threshold scale with V_{CC}.

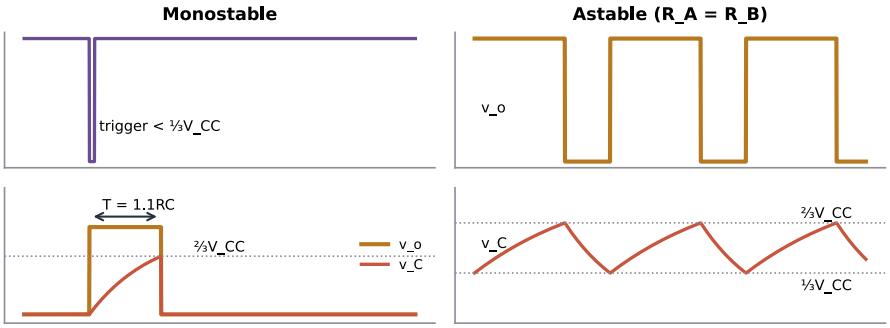


Fig 4.3 Monostable (left) and astable (right) waveforms.

SOLVED EXAMPLE 4.1

Design a 555 monostable for a 1 ms pulse with C = 0.1 µF. What pulse width does R_A = 10 kΩ give?

- $R_A = T/(1.1C) = 1\text{ ms}/(1.1 \times 0.1\ \mu\text{F}) = 9.09\ \text{k}\Omega$
- With 10 kΩ: $T = 1.1 \times 10\text{ k} \times 0.1\ \mu = 1.1\text{ ms}$

$R_A \approx 9.1$ kΩ; 10 kΩ gives 1.1 ms

4.3 555 astable multivibrator

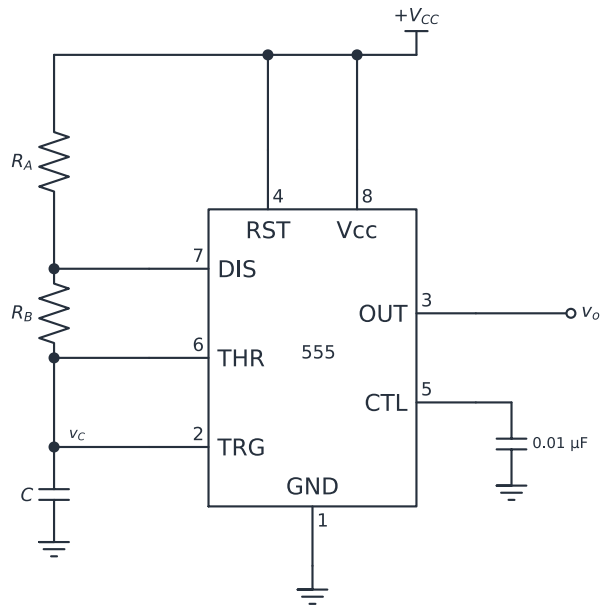


Fig 4.4 555 astable.

Trigger and threshold are tied to the capacitor, so the 555 retriggers itself. C **charges** through $R_A + R_B$ from ⅓ to ⅔V_{CC} (output HIGH), then **discharges** through R_B into pin 7 from ⅔ back to ⅓V_{CC} (output LOW).

Charging between ⅓ and ⅔ of the way to V_{CC} takes RC ln 2 = 0.693RC, both ways.

FORMULA — ASTABLE

$$t_H = 0.693(R_A + R_B)C, \quad t_L = 0.693R_BC$$

$$f = \frac{1.44}{(R_A + 2R_B)C}, \quad D = \frac{R_A + R_B}{R_A + 2R_B}$$

D is always above 50 % because charging goes through both resistors. For 50 % (or less), put a diode across R_B so C charges through R_A alone: $t_H = 0.693R_AC$; then $R_A = R_B$ gives a square wave.

SOLVED EXAMPLE 4.2 · ANALYSIS

An astable has R_A = 2.2 kΩ, R_B = 6.8 kΩ, C = 0.01 µF. Find t_H, t_L, f and D.

- $t_H = 0.693 \times 9\text{ k} \times 0.01\ \mu = 62.4\ \mu s$; $t_L = 0.693 \times 6.8\text{ k} \times 0.01\ \mu = 47.1\ \mu s$
- T = 109.5 µs ⇒ f = 9.13 kHz; D = 62.4/109.5 = 57 %

$t_H \approx 62.4\ \mu s$, $t_L \approx 47.1\ \mu s$, f ≈ 9.13 kHz, D ≈ 57 %

SOLVED EXAMPLE 4.3 · DESIGN

Design a 555 astable for f = 1 kHz, D = 60 %, with C = 0.1 µF.

- T = 1 ms ⇒ t_H = 0.6 ms, t_L = 0.4 ms
- $R_B = t_L/(0.693C) = 0.4\text{ m}/(0.693 \times 0.1\ \mu) = 5.77\ k\Omega$
- $R_A + R_B = 0.6\text{ m}/(0.0693\text{ m}) = 8.66\ k\Omega \Rightarrow R_A = 2.89\ k\Omega$

$R_A \approx 2.9\ k\Omega$, $R_B \approx 5.8\ k\Omega$

4.4 Applications of the 555

Application	Mode	Idea in one line
Missing-pulse detector	mono	each input pulse retriggers (discharges C via an external transistor); if a pulse is missing, T completes and the output drops
Frequency divider	mono	make T slightly longer than (N−1) input periods: it ignores the next N−1 triggers ⇒ output at f/N
Pulse-width modulation	mono	clock on pin 2, modulating signal on pin 5 shifts the ⅔ threshold ⇒ pulse width ∝ signal
Pulse-position modulation	astable	signal on pin 5 changes the period, so each pulse's position shifts
Linear ramp generator	mono	charge C from a constant-current source (PNP) instead of R ⇒ linear ramp
FSK generator	astable	a data bit switches an extra resistor in parallel with R_A ⇒ two frequencies for 1 and 0
Schmitt trigger	—	tie pins 2 and 6 as the input: thresholds ⅓ and ⅔V_{CC}, hysteresis ⅓V_{CC}

INTERVIEW CORNER

Q. Why is the 555's timing independent of supply voltage?

A. The capacitor charges towards V_{CC} and the thresholds are fixed fractions (⅓, ⅔) of V_{CC}, so V_{CC} cancels out of ln(…).

Q. What is pin 5 for?

A. It exposes the ⅔V_{CC} node: decouple it with 0.01 µF to stop noise moving the threshold, or drive it to make PWM/PPM/VCO circuits.

4.5 Voltage-controlled oscillator LM566

A **VCO** produces a frequency proportional to a control voltage — a voltage-to-frequency converter. The 566 charges and discharges C_1 with a constant current set by R_1 and the control voltage V_C; a Schmitt trigger flips the current direction at its two thresholds. Result: a triangle across C_1 (pin 4) and a square wave (pin 3).

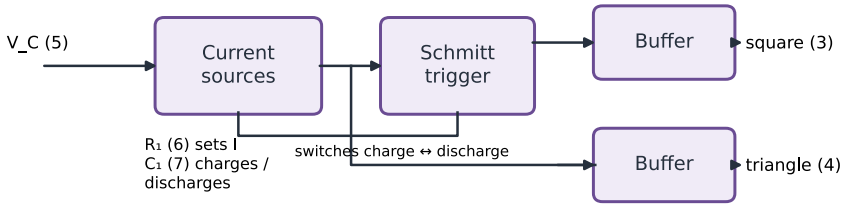


Fig 4.5 LM566 block diagram.

FORMULA — 566

$$f_o = \frac{2(V^+ - V_C)}{R_1 C_1 V^+} \quad \text{with } 0.75V^+ \le V_C \le V^+, \quad 2\ \text{k}\Omega < R_1 < 20\ \text{k}\Omega$$

V_C is set by a divider R_2–R_3 from V^+ (a small capacitor from pin 5 to pin 6 prevents parasitic oscillation). Lowering V_C raises the frequency linearly. Uses: FM and FSK generation, function generators, tone generators — and the heart of a PLL.

SOLVED EXAMPLE 4.4

A 566 runs on $V^+ = 12$ V with $R_1 = 10$ kΩ, $C_1 = 0.001$ µF and $V_C = 10$ V. Find f_o. What is f_o if V_C rises to 10.5 V?

- $f_o = 2(12 - 10)/(10^4 \times 10^{-9} \times 12) = 4/1.2 \times 10^{-4} = 33.3$ kHz
- $V_C = 10.5$ V: $f_o = 2 \times 1.5/(1.2 \times 10^{-4}) = 25$ kHz (both values of V_C lie in 9–12 V √)

33.3 kHz; 25 kHz when $V_C = 10.5$ V

4.6 Phase-locked loop IC 565

A **PLL** is a feedback system that makes a local oscillator (VCO) follow an incoming signal in frequency and phase. Think of a dancer matching a partner's rhythm: if she falls behind, she speeds up; if ahead, slows down — until they move in lock-step.

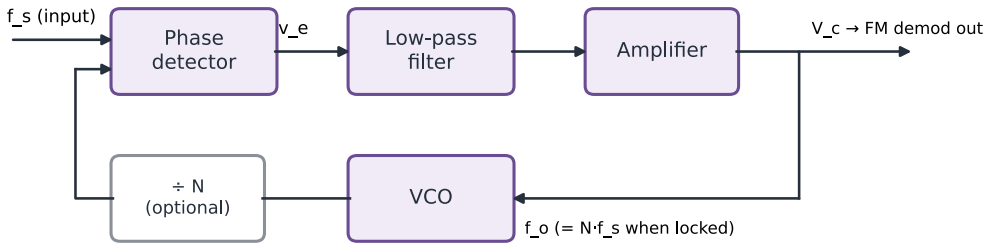


Fig 4.6 PLL block diagram (the ÷N block is added only for frequency multiplication).

- **Phase detector** compares f_s with the VCO output f_o; its output v_e contains a DC part proportional to the phase difference (plus sum-frequency ripple). An analog multiplier or a digital XOR gate does this job.
- **Low-pass filter** removes the ripple and sets the loop's dynamics. **Amplifier** boosts it to the control voltage V_c.
- **VCO** shifts f_o in the direction that reduces the error.

Three states: *free-running* (no input: VCO sits at its free-running f_o), *capture* (input appears; the loop pulls f_o towards f_s), and *lock* ($f_o = f_s$, with a constant small phase difference).

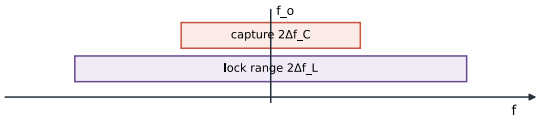


Fig 4.7 Capture range lies inside lock range.

Lock range (tracking range): the band of input frequencies over which a loop that is *already locked* stays locked. **Capture range** (acquisition range): the band over which an *unlocked* loop can acquire lock. Capture is always ≤ lock range, because the LPF attenuates the beat-note the loop needs to pull in.

FORMULA — IC 565

$$f_o = \frac{0.3}{R_1 C_1} \qquad f_L = \pm \frac{8 f_o}{V} \qquad f_C = \pm \frac{1}{2\pi} \sqrt{\frac{2\pi f_L}{(3.6 \times 10^3) C}}$$

V = total supply (+V − (−V)); C = LPF capacitor at pin 7–8 (with the internal 3.6 kΩ resistor)

SOLVED EXAMPLE 4.5 · KTU FAVOURITE

A 565 PLL uses R_1 = 10 kΩ, C_1 = 0.01 µF, LPF capacitor C = 10 µF, supply ±6 V. Find f_o, lock range and capture range.

- $f_o = 0.3/(10^4 \times 10^{-8}) = 3$ kHz
- V = 12 V ⇒ $f_L = \pm 8 \times 3000/12 = \pm 2$ kHz (lock range 1–5 kHz)
- $f_C = \pm(1/2\pi)\sqrt{(2\pi \times 2000/(3600 \times 10 \times 10^{-6}))} = \pm(1/2\pi)\sqrt{349\,066} = \pm(590.8/6.283) = \pm 94$ Hz

f_o = 3 kHz, f_L = ±2 kHz, f_C ≈ ±94 Hz

4.7 PLL applications: FM demodulation, frequency multiplication

- **FM demodulation:** feed the FM signal to a locked PLL. To track the changing input frequency, the VCO control voltage V_c must change in proportion to the frequency deviation — so V_c (after the amplifier) *is* the demodulated audio. The VCO's linearity sets the distortion.
- **Frequency multiplication / synthesis:** insert a ÷N counter between the VCO and the phase detector. Lock forces $f_o/N = f_s$, so $\mathbf{f_o = N \cdot f_s}$. With a programmable N and a crystal reference, this is the frequency synthesiser in every radio and the clock multiplier inside every processor.
- **FSK demodulation:** the two input tones give two levels of V_c — a comparator turns them into 1s and 0s.
- Others: AM detection, tracking filters, clock recovery in serial links, motor-speed control.

INTERVIEW CORNER

Q. Why is capture range smaller than lock range?
A. Before lock, the phase-detector output is an AC beat at $|f_s - f_o|$; the LPF attenuates it, so only near-by frequencies produce enough DC to pull the VCO in. Once locked the error is DC and passes freely.

Q. Effect of the loop filter bandwidth?
A. Narrow filter: better noise rejection, smaller capture range, slower acquisition. Wide filter: fast and large capture, but more jitter.

Q. How do you get 100 MHz from a 10 MHz crystal?
A. PLL with a ÷10 counter in the feedback path: the VCO locks at 10 × 10 MHz.

4.8 DAC specifications and weighted-resistor DAC

A **DAC** turns an n-bit number into a proportional voltage or current: $v_o = K \cdot (b_{n-1}2^{n-1} + \ldots + b_1 2 + b_0)$. Each bit contributes a weight twice that of the bit to its right — the whole design problem is generating these binary weights accurately.

Specification	Meaning
Resolution	smallest output step = 1 LSB = $V_{FS}/2^n$ (often quoted as n bits or $1/2^n$); 8-bit, 5 V ⇒ 19.5 mV
Accuracy	difference between actual and ideal output (in % FS or LSB)
Linearity	maximum deviation of the transfer staircase from a straight line (ideally < ½ LSB)
Monotonicity	output never decreases when the input code increases
Settling time	time to settle within ±½ LSB after a full-scale code change
Offset / gain error	output for code 0; deviation of the full-scale slope

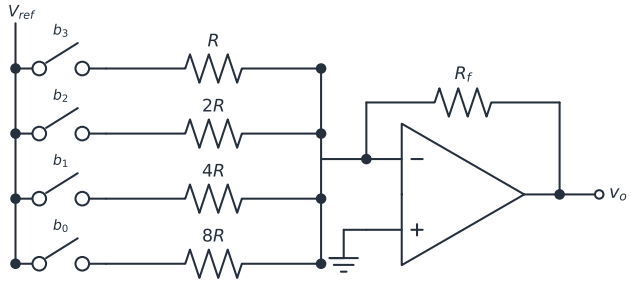


Fig 4.8 4-bit binary-weighted resistor DAC.

Each bit's switch connects V_{ref} to a resistor whose value doubles for each less-significant bit. The summing amplifier adds the currents V_{ref}/R, $V_{ref}/2R$, ...:

FORMULA

$$v_o = -V_{ref}\frac{R_f}{R}\left(b_3 + \frac{b_2}{2} + \frac{b_1}{4} + \frac{b_0}{8}\right)$$

Drawback: resistor spread of $2^{n-1}:1$ (128 : 1 for 8 bits, 1 : 2048 for 12 bits) — wide-ranging precision values are very hard to make on a chip.

4.9 R-2R and inverted R-2R ladder DACs

The R-2R ladder uses just **two** resistor values. Its magic: looking into any node towards the LSB end you always see 2R ∥ 2R = R in series with R... = 2R again. So at each node the current (or voltage) splits exactly in half.

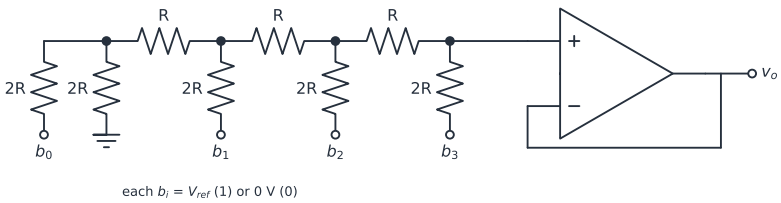


Fig 4.9 R-2R ladder (voltage mode) with buffer.

Voltage mode (Fig 4.9): each bit drives its 2R leg with V_{ref} or 0 V; by superposition each bit's contribution is halved at every node it passes, giving $v_o = V_{ref}(b_3/2 + b_2/4 + b_1/8 + b_0/16)$. The output resistance is always R.

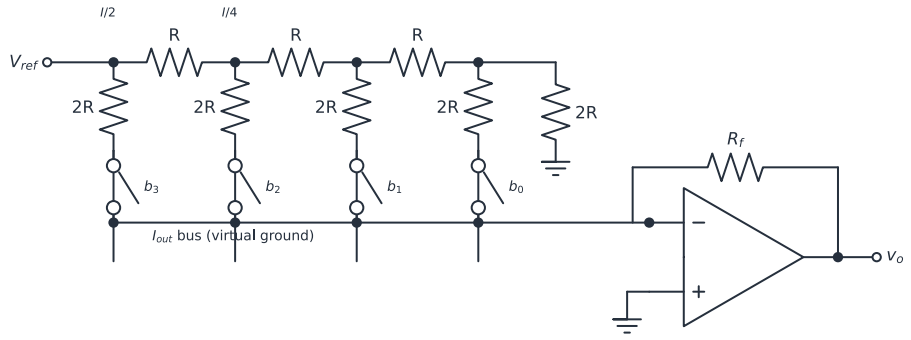


Fig 4.10 Inverted (current-mode) R-2R ladder: each switch steers its current to the virtual-ground bus or to ground.

Inverted R-2R (Fig 4.10): V_{ref} feeds the ladder and the switches sit at the bottom of the 2R legs. Both switch positions are at 0 V (true ground or virtual ground), so the ladder's node voltages **never change** — the currents I/2, I/4, I/8, I/16 ($I = V_{ref}/R$) flow constantly and are just steered. No stray capacitance needs charging, so it is the **fastest** and most popular structure.

FORMULA — R-2R

$$\text{Inverted R-2R: } v_o = -R_f \frac{V_{ref}}{R}\left(\frac{b_3}{2} + \frac{b_2}{4} + \frac{b_1}{8} + \frac{b_0}{16}\right) = -V_{ref}\frac{R_f}{R}\cdot\frac{D}{2^n}$$

SOLVED EXAMPLE 4.6 · DAC OUTPUTS

(a) A 4-bit weighted-resistor DAC has V_{ref} = 5 V, R_f = R. Find v_o for 1011 and the step size. (b) A 4-bit inverted R-2R DAC has V_{ref} = 10 V, R_f = R. Find v_o for 1011 and the resolution. (c) Resolution of an 8-bit, 5 V DAC?

- (a) $v_o = -5(1 + 0 + 0.25 + 0.125) = -6.875$ V; LSB = 5/8 = 0.625 V
- (b) D = 11 ⇒ $v_o = -10 \times 11/16 = -6.875$ V; resolution = 10/16 = 0.625 V
- (c) 5/256 = 19.5 mV (0.39 %)

(a) −6.875 V, 0.625 V (b) −6.875 V, 0.625 V (c) 19.5 mV

4.10 ADC specifications and flash ADC

An **ADC** does the reverse: it measures an analog voltage and outputs the nearest n-bit code. Key specifications: **resolution** (n bits, LSB = $V_{FS}/2^n$), **quantisation error** (±½ LSB, unavoidable), **conversion time**, **accuracy** and **missing codes**/linearity. The input must stay still during conversion — hence the sample-and-hold of Module 2.

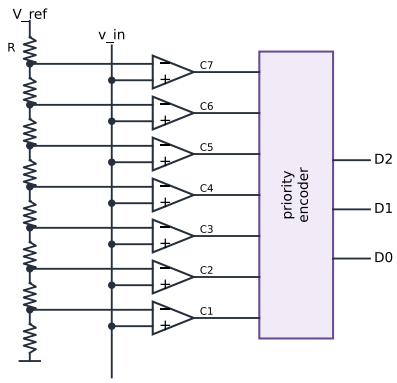


Fig 4.11 3-bit flash ADC.

Flash (parallel) ADC. A resistor ladder makes $2^n - 1$ reference levels; one comparator per level compares v_{in} with all of them *simultaneously*. Comparators below v_{in} output 1, those above output 0 — a **thermometer code** — which a priority encoder turns into binary.

+ fastest possible (one comparator delay, GHz rates) — used in oscilloscopes, radar, video. − comparator count doubles with every bit: 8 bits need 255, so power and area explode.

SOLVED EXAMPLE 4.7 · FLASH ADC

A 3-bit flash ADC with V_{ref} = 8 V uses eight equal resistors. Find the comparator thresholds and the output for v_{in} = 5.3 V. How many comparators would 8 bits need?

- Thresholds: 1, 2, 3, 4, 5, 6, 7 V (C1…C7)
- 5.3 V exceeds 1–5 V ⇒ C1–C5 = 1, C6, C7 = 0 ⇒ thermometer 0011111 ⇒ binary 101
- 8 bits: $2^8 - 1$ = 255 comparators

Output 101 (= 5); 255 comparators for 8 bits

4.11 Counter-type and successive-approximation ADCs

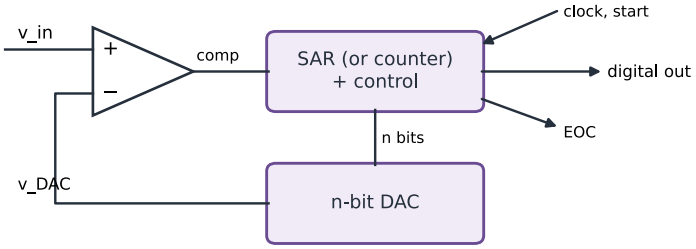


Fig 4.12 Feedback ADC: a DAC's output is compared with v_{in}; the logic is a counter or a SAR.

Counter (digital-ramp) type. Reset the counter; count up one step per clock; the DAC output climbs as a staircase. When it passes v_{in} the comparator flips, counting stops and the count is the answer. Simple, but slow and variable: up to $2^n - 1$ clocks (255 µs for 8 bits at 1 MHz). The **tracking** version uses an up/down counter that follows v_{in}.

Successive approximation (SAR). A binary search — like guessing a number between 0 and 255 by asking "is it ≥ 128?", then "≥ 192?", … Starting from the MSB, the SAR sets each bit to 1, lets the DAC produce that trial voltage, and keeps the bit only if the trial ≤ v_{in}. **Exactly n clocks** per conversion, whatever v_{in} is (8 µs for 8 bits at 1 MHz). It is the most widely used ADC (microcontrollers, data acquisition).

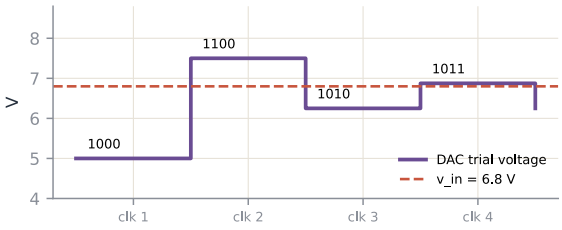


Fig 4.13 SAR trials for Example 4.8.

SOLVED EXAMPLE 4.8 · SAR STEP-BY-STEP

A 4-bit SAR ADC has full scale 10 V (LSB = 0.625 V). Convert v_{in} = 6.8 V.

- 1000 → 5.0 V ≤ 6.8 ⇒ keep 1
- 1100 → 7.5 V > 6.8 ⇒ clear ⇒ 1000
- 1010 → 6.25 V ≤ 6.8 ⇒ keep
- 1011 → 6.875 V > 6.8 ⇒ clear ⇒ 1010

1010 (6.25 V); error 0.55 V < 1 LSB

4.12 Dual-slope ADC

The dual-slope ADC converts voltage to **time** and counts the time. **Phase 1:** integrate v_{in} for a fixed time T_1 (2^n clock counts) — the integrator output reaches a value proportional to v_{in}. **Phase 2:** switch to $-V_{ref}$ and count clocks until the integrator returns to zero. The fall slope is fixed, so the time T_2 is proportional to v_{in}.

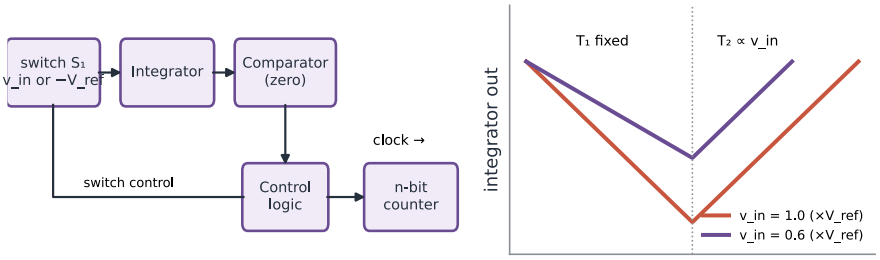


Fig 4.14 Dual-slope ADC and its integrator waveform for two inputs.

End of phase 1: $v_1 = -v_{in}T_1/RC$. Phase 2 removes it: $V_{ref}T_2/RC = v_{in}T_1/RC$

FORMULA — DUAL SLOPE

$$T_2 = T_1\frac{v_{in}}{V_{ref}} \Rightarrow N_2 = 2^n\frac{v_{in}}{V_{ref}}$$ (R, C and clock frequency all cancel)

Why everyone's multimeter uses it: accuracy depends only on V_{ref}; integration averages noise; choosing T_1 = a multiple of 20 ms makes 50 Hz hum integrate to exactly zero. **Price:** slow (tens of ms per reading).

SOLVED EXAMPLE 4.9 · DUAL SLOPE

A dual-slope ADC has T_1 = 1000 counts, V_{ref} = 1 V and a 50 kHz clock. Find the count and total conversion time for v_{in} = 0.4 V. Does it reject 50 Hz?

- $N_2 = 1000 \times 0.4/1 = 400$ counts
- $T_1 = 1000/50$ kHz = 20 ms, $T_2 = 400/50$ kHz = 8 ms ⇒ total ≈ 28 ms
- T_1 = 20 ms = exactly one 50 Hz period ⇒ hum averages to zero ✓

400 counts, ≈ 28 ms, 50 Hz rejected

INTERVIEW CORNER

Q. Why is the inverted R-2R faster than the voltage-mode ladder?
A. Its node voltages stay constant (switches toggle between ground and virtual ground), so no parasitic capacitance charges or discharges on a code change.

Q. What is quantisation error and can it be removed?
A. The difference between v_{in} and the code's voltage, up to ±½ LSB. It can only be reduced (more bits), never removed.

Q. Why does a SAR ADC need a sample-and-hold?
A. The binary search assumes v_{in} is constant for all n trials; a moving input gives wrong bits.

✓ Module 4 — Quick revision

555 thresholds: $\frac{2}{3}V_{CC}$ (threshold → reset), $\frac{1}{3}V_{CC}$ (trigger → set)

Monostable: $T = 1.1R_AC$ (= RC ln 3)

Astable: $t_H = 0.693(R_A+R_B)C$, $t_L = 0.693R_BC$

Astable: $f = 1.44/((R_A+2R_B)C)$, $D = (R_A+R_B)/(R_A+2R_B) > 50\ \%$

566: $f_o = 2(V^+ - V_C)/(R_1C_1V^+)$, $0.75V^+ \le V_C \le V^+$

565: $f_o = 0.3/R_1C_1$; $f_L = \pm 8f_o/V$; $f_C = \pm(1/2\pi)\sqrt{(2\pi f_L/(3.6k\cdot C))}$

PLL: PD → LPF → amp → VCO; capture ≤ lock. FM demod = V_c; ÷N ⇒ $f_o = Nf_s$

DAC LSB = $V_{FS}/2^n$; weighted: spread 2^{n-1}:1

R-2R: $v_o = V_{ref}\cdot D/2^n$ ($\times R_f/R$, inverted); inverted ladder fastest

Flash: $2^n - 1$ comparators, 1 clock

Counter: ≤ $2^n - 1$ clocks. **SAR:** n clocks

Dual slope: $T_2 = T_1v_{in}/V_{ref}$; independent of RC, clock; rejects 50 Hz if T_1 = k × 20 ms

Quantisation error: ±½ LSB

Practice — short answer (3 marks)

1. Draw the internal block diagram of the 555.
2. Design a 555 monostable for 5 ms with C = 1 µF. [4.55 kΩ]
3. How is 50 % duty obtained in a 555 astable?
4. Define lock range and capture range.
5. How does a PLL act as a frequency multiplier?
6. Define resolution, settling time and monotonicity of a DAC.
7. Why is the R-2R ladder preferred to the weighted-resistor DAC?
8. Compare flash, SAR and dual-slope ADCs.

Practice — long answer (9 / 14 marks)

1. Explain the 555 astable with waveforms; derive f and D. Design for 2 kHz, D = 70 % with C = 0.01 µF. [$R_B \approx 21.6$ kΩ, $R_A \approx 28.9$ kΩ]
2. Explain the LM566 VCO and derive the relation between f_o and V_C.
3. Draw the PLL block diagram; explain lock and capture. For 565 with R_1 = 15 kΩ, C_1 = 0.01 µF, C = 1 µF, ±10 V find f_o, f_L, f_C. [2 kHz, ±800 Hz, ±188 Hz]
4. Explain the inverted R-2R ladder DAC and derive its output.
5. Explain the successive-approximation ADC with an example conversion.
6. Explain the dual-slope ADC and derive T_2; list its advantages.

⚡ Placement & GATE rapid-fire

Cover the answer, say yours out loud in one or two sentences, then check. Interviewers at core companies care more about the *reason* than the formula. Starred (★) ones are deliberately tricky.

1. **List the ideal op-amp properties.** A, R_i, bandwidth, CMRR, slew rate → ∞; R_o, offsets, bias currents, drift → 0.
2. **Why is the input stage a differential amplifier?** It amplifies the difference and rejects common-mode noise and drift, and allows direct (DC) coupling.
3. **What limits CMRR in practice?** Mismatch between the two halves and the finite resistance of the tail current source.
4. **Why current mirrors instead of resistors in ICs?** Small area, good matching and temperature tracking, and high output resistance.
5. **What does a Widlar source buy you?** Very small currents (µA) with modest resistors, plus higher output resistance.
6. **What causes slew-rate limiting?** The limited current available to charge the internal compensation capacitor: SR = I/C_c.
7. **741 (f_t = 1 MHz) as a non-inverting amp of gain 101 — bandwidth?** 1 MHz/101 ≈ 9.9 kHz.
8. **Largest undistorted 10 kHz sine from a 741?** SR/(2πf) = $0.5 \times 10^6/(2\pi \times 10^4)$ ≈ 7.96 V peak.
9. **What is phase margin?** 180° plus the loop phase at the frequency where loop gain = 1; ≥ 45° means stable with little ringing.
10. **Why is the 741 "internally compensated"?** A built-in 30 pF dominant-pole capacitor makes it stable even at unity gain.
11. **Give two ways to remove output offset.** Null pot on pins 1–5 for V_{io}; $R_{comp} = R_1 \parallel R_f$ on the + input for bias current.
12. **What is a virtual ground?** The inverting node held at 0 V by feedback, though not connected to ground and drawing no current.
13. **Why is the inverting amp's input resistance only R_1?** The source sees R_1 going to a virtual ground.
14. **★ R_f accidentally connected to the + input?** That is positive feedback: the output latches at $\pm V_{sat}$ (it becomes a Schmitt trigger).
15. **★ Inverting amp, gain −10, input 2 V, supplies ±12 V?** It cannot give −20 V; it saturates near −10.5 to −11 V.
16. **Why use an instrumentation amp rather than a difference amp?** High, equal input impedances, high CMRR, and gain set by one resistor R_G.
17. **1 V DC into an integrator with R = 1 MΩ, C = 1 µF?** The output ramps at −1 V/s until it saturates.
18. **Why does a practical integrator have R_f?** It limits the DC gain so offsets don't drive the output into saturation.
19. **Why is an ideal differentiator avoided?** Its gain rises with frequency, so it amplifies noise and can oscillate.
20. **What does a precision rectifier achieve?** Rectification of millivolt signals: the effective diode drop becomes 0.7 V/A.
21. **Why use a dedicated comparator (LM311) rather than a 741?** No compensation capacitor so it is much faster, has an open-collector output for logic levels, and recovers quickly from saturation.
22. **Hysteresis of an inverting Schmitt with R_1 = 1 k, R_2 = 9 k, ±10 V?** $V_H = 2 \times 10 \times 1/10 = 2$ V.
23. **Log-amp output change per decade of input?** $V_T \ln 10 \approx 60$ mV at room temperature.
24. **Peak detector vs sample-and-hold?** A peak detector holds the maximum value; an S/H holds whatever value exists at the hold command.
25. **Why doesn't an oscillator's amplitude grow forever?** Amplifier saturation, or an AGC element (lamp, thermistor, diodes), reduces the gain until $|A\beta| = 1$.
26. **Wien bridge with R = 10 kΩ, C = 10 nF: frequency?** $1/(2\pi RC) \approx 1.59$ kHz.

27. **Why does an RC phase-shift oscillator need gain 29?** The 3-section network attenuates by 1/29 at the 180° frequency.
28. **Butterworth vs Chebyshev?** Butterworth: flat pass band, gentler roll-off. Chebyshev: ripple in the pass band, steeper roll-off.
29. **Filter order needed for −60 dB/decade?** Third order (−20 dB/dec per order).
30. **What does the Q of a band-pass filter mean?** Q = f_c/BW, i.e. selectivity: high Q means a narrow, sharp peak.
31. **Where is an all-pass filter used?** Phase or delay equalisation, and as a phase shifter. Its gain is flat at all frequencies.
32. **Line vs load regulation?** Line: ΔV_o for a change in V_{in}. Load: ΔV_o from no load to full load.
33. **Linear vs switching regulator?** Linear: quiet and simple, but wastes $(V_{in} - V_o)I$ as heat. Switching: 80–95 % efficient, but noisier.
34. **Why foldback limiting?** It cuts the short-circuit current well below I_{limit}, so the pass transistor survives a short.
35. **Why is 555 timing independent of V_{CC}?** The thresholds (⅓, ⅔V_{CC}) and the charging target both scale with V_{CC}.
36. **555 astable with $R_A = R_B = 10$ kΩ, $C = 0.1$ µF?** f = 1.44/(30 k × 0.1 µ) = 480 Hz, D = 2/3 ≈ 67 %.
37. **What happens when the 555 reset pin is pulled low?** The output is forced low at once and the discharge transistor turns on, overriding everything else.
38. **PLL: lock range vs capture range?** Lock: range over which an already-locked loop stays locked. Capture: range over which it can acquire lock. Capture ≤ lock.
39. **How does a PLL multiply frequency?** Put a ÷N counter in the feedback path, so $f_o = N \cdot f_{in}$.
40. **Why is the PLL control voltage the FM-demodulated output?** To follow the input, V_c must be proportional to its instantaneous frequency deviation.
41. **Why is the inverted R-2R the fastest DAC?** Its node voltages never change, so no stray capacitances need to charge.
42. ★ **What makes a DAC non-monotonic?** Weight mismatch at a major carry (e.g. 0111 → 1000) larger than 1 LSB, so the output steps down.
43. **LSB of a 10-bit, 5 V ADC?** 5/1024 ≈ 4.88 mV.
44. **Comparators in a 4-bit flash ADC?** $2^4 - 1 = 15$.
45. **Conversion time of a 12-bit SAR at 1 MHz?** About 12 µs (n clocks), independent of the input.
46. **Why do multimeters use dual-slope ADCs?** Accuracy depends only on V_{ref}, noise averages out, and $T_1 = 20$ ms rejects 50 Hz hum.
47. ★ **Ideal SNR of an 8-bit ADC?** 6.02n + 1.76 ≈ 49.9 dB (full-scale sine, quantisation noise only).

Σ Formula sheet

Module 1 — op-amp internals	Formula
Diff-amp DC	$I_E = (V_{EE} - V_{BE})/(2R_E + R_{in}/\beta)$; $V_{CE} = V_{CC} + V_{BE} - I_C R_C$; $r_e = V_T/I_E$
Diff-amp AC	$A_d = R_C/r_e$ (balanced), $R_C/2r_e$ (unbalanced); $R_i = 2\beta r_e$; $A_{cm} = R_C/(r_e + 2R_E)$; CMRR ≈ R_E/r_e
Output with CMRR	$v_o = A_d v_d + A_{cm} v_{cm}$, $v_{cm} = (v_1 + v_2)/2$; CMRR(dB) = 20 log(A_d/A_{cm})
Current source / mirrors	$I_T = (V_{EE}R_2/(R_1+R_2) - V_{BE})/R_E$; basic $I_O = I_{ref}/(1 + 2/\beta)$; Wilson $I_O = I_{ref}(1 - 2/(\beta^2+2\beta+2))$; Widlar $I_O R_E = V_T \ln(I_{ref}/I_O)$
Output offset	$V_{oo} = (1 + R_f/R_1)V_{io} + R_f I_{io}$ (with $R_{comp} = R_1 \parallel R_f$); uncompensated bias term $I_B R_f$

Slew rate, GBW, VTC	SR = I/C_c; $f_{max} = SR/(2\pi V_m)$; $A_{CL}f_{CL} = f_t$; linear range $\pm V_{sat}/A$

Module 2 — feedback & applications	**Formula**
Basic amplifiers, converters	Inverting $-R_f/R_1$; non-inverting $1 + R_f/R_1$; follower 1; V→I $i_L = v_{in}/R_1$; I→V $v_o = -i_{in}R_f$
Negative feedback	$A_F = A/(1 + A\beta)$; $R_{iF} = R_i(1 + A\beta)$; $R_{oF} = R_o/(1 + A\beta)$; $f_F = f_o(1 + A\beta)$
Summer / difference / IA	$-R_f\Sigma v_k/R_k$; $(R_2/R_1)(v_2 - v_1)$; $(R_4/R_3)(1 + 2R/R_G)(v_2 - v_1)$
Integrator	$-(1/R_1C_f)\int v_{in}dt$; $f_a = 1/2\pi R_fC_f$, $f_b = 1/2\pi R_1C_f$
Differentiator	$-R_fC_1dv_{in}/dt$; $f_a = 1/2\pi R_fC_1$, $f_b = 1/2\pi R_1C_1 = 10–20f_a$, $R_1C_1 = R_fC_f$
Schmitt triggers	Inverting $\pm R_1V_{sat}/(R_1+R_2)$ (with V_{ref}: $(R_2V_{ref} \pm R_1V_{sat})/(R_1+R_2)$); non-inverting $\pm(R_1/R_2)V_{sat}$
Log / antilog / S&H	$-V_T\ln(v_{in}/RI_S)$; $-RI_Se^{v_{in}/V_T}$; droop $= I_{leak}/C_H$

Module 3 — oscillators, filters, 723	**Formula**
Oscillators	$\lvert A\beta\rvert = 1$, $\angle = 0°$; RC phase-shift $f = 1/(2\pi\sqrt{6}RC)$, $A \geq 29$; Wien $f = 1/(2\pi RC)$, $A = 3$
Generators	Astable $T = 2RC\ln((1+\beta)/(1-\beta))$, $\beta = R_1/(R_1+R_2)$; triangle $f = R_2/(4R_1RC)$, $V_{pp} = 2(R_1/R_2)V_{sat}$; monostable $T = RC\ln((1 + V_D/V_{sat})/(1 - \beta))$
Butterworth LPF/HPF	$f_c = 1/2\pi RC$; 1st order $A_F = 1 + R_f/R_1$; 2nd order $A_F = 1.586$; $\lvert H\rvert = A_F/\sqrt{(1 + (f/f_H)^{2n})}$; scaling $R' = R\,f_c/f_c'$
Band-pass / reject / all-pass	$f_c = \sqrt{(f_Lf_H)}$, $Q = f_c/BW$; MFB $R_1 = Q/(2\pi f_cCA_F)$, $R_2 = Q/(2\pi f_cC(2Q^2 - A_F))$, $R_3 = Q/(\pi f_cC)$; notch $f_N = 1/2\pi RC$; all-pass $\varphi = -2\tan^{-1}(2\pi fRC)$
IC 723	$V_{ref} \approx 7.15$ V; low $V_o = V_{ref}R_2/(R_1+R_2)$; high $V_o = V_{ref}(1 + R_1/R_2)$; $R_{sc} = V_{sense}/I_{limit}$; foldback $I_{sc} = V_{sense}(R_3+R_4)/(R_{sc}R_4)$

Module 4 — 555, VCO, PLL, converters	**Formula**
555	Monostable $T = 1.1RC$; astable $t_H = 0.693(R_A+R_B)C$, $t_L = 0.693R_BC$, $f = 1.44/((R_A+2R_B)C)$, $D = (R_A+R_B)/(R_A+2R_B)$
566 / 565	$f_o = 2(V^+ - V_C)/(R_1C_1V^+)$; 565 $f_o = 0.3/R_1C_1$, $f_L = \pm 8f_o/V$, $f_C = \pm(1/2\pi)\sqrt{(2\pi f_L/(3.6\ k\Omega\cdot C))}$
DACs	LSB $= V_{FS}/2^n$; weighted $-V_{ref}(R_f/R)\Sigma b_i2^{i-(n-1)}$; R-2R $V_{ref}\cdot D/2^n$ (× $-R_f/R$ for inverted)
ADCs	Quantisation error ±½ LSB; flash $2^n - 1$ comparators; dual slope $T_2 = T_1v_{in}/V_{ref}$; SNR ≈ 6.02n + 1.76 dB. Conversion time: flash 1 clock (fastest), counter ≤ $2^n - 1$, SAR n, dual slope ≤ $2{\cdot}2^n$ (slowest, most accurate)

☾ Night before the exam

Breathe. You don't need everything — you need the patterns below and the confidence to apply the two golden rules. Read this page slowly once, then sleep.

Two golden rules (negative feedback): $v_+ = v_-$ and no input current. They solve 80 % of Module 2–3 circuits.

Diff amp in 3 lines: $r_e = 26\ mV/I_E$; $A_d = R_C/r_e$ (half if unbalanced); CMRR ≈ R_E/r_e ⇒ use a current source.

Mirrors: basic $2/\beta$ error; Wilson $2/\beta^2$ error + high R_o; Widlar = µA currents with small R.

741 numbers: 2×10^5, 2 MΩ, 75 Ω, 0.5 V/µs, 1 MHz, 90 dB, 2 mV, 80 nA, 20 nA.

Speed limits: small signal → GBW (f_t/A_{CL}); large signal → slew rate ($SR/2\pi V_m$).

Integrator ⇄ differentiator: swap R and C; practical versions add R_f (integrator) or R_1, C_f (differentiator).

Schmitt: positive feedback ⇒ two thresholds; always draw the loop with arrows.

Oscillators: phase-shift (inverting amp, 1/29, $\sqrt{6}$), Wien (non-inverting, 1/3, 1/RC).

Filter design recipe: pick f_c, pick C, $R = 1/2\pi f_c C$, set A_F (1.586 for 2nd-order Butterworth).

723: V_{ref} 7.15 V; below 7 V divide V_{ref}, above 7 V divide V_o; $R_{sc} = 0.5\ V/I_{lim}$.

555: ⅓ and ⅔V_{CC}; mono 1.1RC; astable 0.693 with (R_A+R_B) for high, R_B for low.

PLL: PD → LPF → amp → VCO; capture ≤ lock; ÷N ⇒ ×N; V_c = FM output.

Converters: R-2R only two values; inverted ladder fastest. Flash fastest; SAR n clocks; dual slope most accurate & slow.

Units sanity: RC in seconds (kΩ × µF = ms); always state units and round to 3 significant figures.

EXAM-HALL STRATEGY

- For every derivation: draw the circuit, mark currents and node voltages, state the assumption (ideal op-amp, virtual ground), then write the equations. Neat diagrams earn marks on their own.
- For design questions: write the formula, choose standard values (C first), compute, and box the final values with units. Mention practical tweaks (pot for R_f, nearest standard resistor).
- Waveform questions: draw input and output on the same time axis with levels marked ($\pm V_{sat}$, thresholds, ⅓/⅔V_{CC}).
- Stuck? Attempt every part — KTU marks steps. You have got this.